Remote Sensing of Environment 253 (2021) 112174

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

A million kernels of truth: Insights into scalable satellite maize yield mapping and yield gap analysis from an extensive ground dataset in the US Corn Belt

Jillian M. Deines [a,*], Rinkal Patel [b], Sang-Zi Liang [c], Walter Dado [a], David B. Lobell [a]

[a] Department of Earth System Science, Center on Food Security and the Environment, Stanford University, United States of America
[b] Granular, A Corteva Agriscience™ Company, United States of America
[c] Corteva Agriscience™, United States of America

ARTICLE INFO

Keywords:
Crop yields
US Corn Belt
Landsat
Agricultural monitoring

ABSTRACT

Crop yield maps estimated from satellite data increasingly are used to understand drivers of yield trends and variability, yet satellite-derived regional maps are rarely compared with location-specific yields due to the difficulty of acquiring sub-field ground truth data at scale. In commercial agricultural systems, combine harvesters with onboard yield monitors collect real-time yield data during harvest with high spatial resolution, generating a large ground dataset. Here, we leveraged a yield monitor dataset of over one million maize field observations across the United States Corn Belt from 2008 to 2018 to evaluate the Scalable Crop Yield Mapper (SCYM). SCYM uses region-specific crop model simulations and climate data to interpret vegetation indices from satellite observations, thus enabling efficient sub-field yield estimation across large regions and multiple years without reliance on ground data calibration. We used the ground dataset to compare alternative SCYM model implementations, define minimum required satellite observation criteria, and evaluate the sensitivity of satellite-based yield estimates to on-the-ground variation in management, soil, and annual weather. We found that smoothing annual time series data with harmonic regression increased 30 m pixel-level accuracy from $r^2 = 0.31$ to 0.40 and reduced dependency on specific satellite observation timing, enabling robust yield estimation on 97% of annual maize area using only Landsat data. Agreement improved as the assessment was aggregated to field-level ($r^2 = 0.45$) and county-level ($r^2 = 0.69$) scales, demonstrating the need for fine-resolution ground truth data to better assess sub-field level accuracy in high resolution products. We found that SCYM and ground data showed a similar yield response to management and environmental variation, particularly capturing linear and nonlinear responses to sowing density, soil water holding capacity, and growing season precipitation. However, sensitivity to factors like soil quality and planting date was muted for SCYM estimates compared to ground-based yields. Random forest models were able to match SCYM performance when trained on at least 1000 ground observations, but performed poorly when tested on years and locations not represented in the training data. Our results demonstrate that satellite yield maps can provide much-needed information on multidecadal yield trends and inform yield gap analyses.

1. Introduction

Satellite monitoring of agricultural systems can support efforts addressing food production and sustainability around the world by providing information with consistent methodology and frequency (Weiss et al., 2020). Monitoring of crop yields in particular can inform management interventions, food security efforts, research priorities, and commodity markets (Carletto et al., 2015; Hoffman et al., 2015; Jain et al., 2019; Lobell et al., 2009; Maestrini and Basso, 2018; Van Ittersum et al., 2013). Ongoing advances in the accessibility and application of satellite data for retrospective crop yield mapping increasingly enable agricultural monitoring at high spatial and temporal frequencies on decadal scales. This provides opportunities for detailed detection of multitemporal yield heterogeneity among and within fields that can provide greater insights into factors affecting yield performance than governmental statistics (Diker et al., 2004; Lobell, 2013), which tend to

* Corresponding author at: Department of Earth Systems Science, Stanford University, 473 Via Ortega Dr. Room 349, Stanford, CA 94305, United States of America.
E-mail address: jillian.deines@gmail.com (J.M. Deines).

https://doi.org/10.1016/j.rse.2020.112174
Received 13 April 2020; Received in revised form 1 October 2020; Accepted 2 November 2020
Available online 15 November 2020
0034-4257/© 2020 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

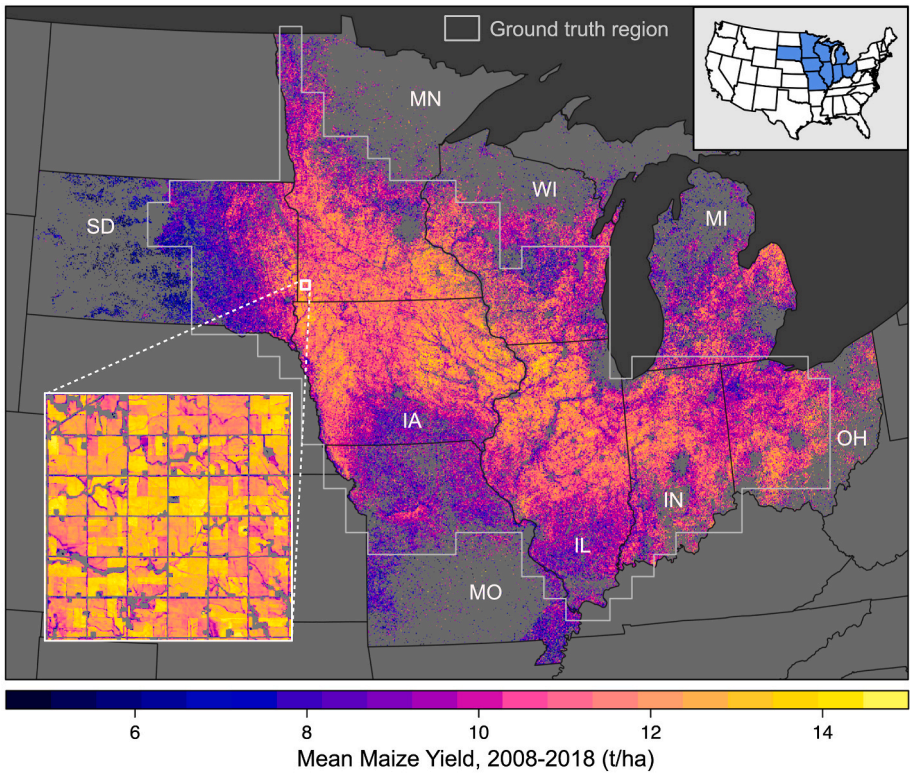


Fig. 1. Study area and mean 2008–2018 yields. The extent of the ground truth data from yield monitors used to evaluate alternative yield mapping approaches is outlined in gray (see also Fig. 2). Mean yields for the larger nine-state study region were generated by applying the preferred SCYM model to Landsat satellite data.

be aggregated to coarse-scale political units and often unavailable in data-sparse regions. Since the first applications of satellite imagery to crop yield estimation in the 1970s (MacDonald and Hall, 1980), efforts have included lower-resolution (250–1000 m) maps across country-to-global scales (e.g., Becker-Reshef et al., 2010; Franch et al., 2019; Kamir et al., 2020; Whitcraft et al., 2015) and higher-resolution (<30 m) yield maps generally restricted to local scales due to time or computational costs, limited ground data, or imagery limitations (e.g, Gao et al., 2018; Kang and Özdogan, 2019; Lambert et al., 2018). With recent improvements in imagery accessibility and computational power (Azzari and Lobell, 2016; Wulder et al., 2018), high resolution, sub-field scale estimates are now increasingly deployed at regional scales across multiple years (e.g, Jin et al., 2019; Skakun et al., 2019).

The reliability of these estimates, however, are often unclear because of the lack of reliable, spatially explicit ground truth data with which to evaluate them. Extensive, high-resolution ground yields are difficult to acquire due to costs and privacy concerns. Instead, many studies necessarily aggregate yields to larger political units for comparison with government statistics, leaving uncertainty about their ability to accurately capture subfield variability (Hunt et al., 2019; Jin et al., 2017; Kang and Özdogan, 2019). Due to averaging effects, aggregated assessments are likely an upper bound on accuracy compared to field-level or pixel-based comparisons.

With the rise of precision agriculture in large commercial systems, an increasing number of farmers now employ combine harvesters equipped with yield monitor technology. Yield monitors use high-accuracy GPS and on-board sensors to estimate grain weight and moisture to produce fine resolution (~5 m or less) yield maps during crop harvest. These maps can then be used by field managers to assess yield potentials or create management zones, enabling them to target nutrient or water inputs or other interventions more efficiently (Kharel et al., 2019). Relatively small yield monitor datasets of two to several hundred fields have been used to assess within-field yield performance such as yield stability due to topography, soils, and shallow groundwater availability (Maestrini and Basso, 2018; Zipper et al., 2015). Other efforts have used these yield monitor data to train empirical satellite models. For example, Hunt et al. (2019) used yield monitor data from 39 wheat fields to calibrate a satellite model to map wheat yields at 10 m resolution within a single Sentinel-2 image tile in the United Kingdom, and Lai et al. (2018) developed a linear mixed-effects model from 60 yield monitor field maps to predict within-field wheat yields on 17 farms in Australia from 2001 to 2016 using Landsat data.

The increased availability of detailed ground datasets will undoubtedly foster further advances in using machine learning for satellite-based crop yield prediction. At the same time, large ground truth datasets such as yield monitors are still difficult to acquire and are typically concentrated in commercial systems or recent years, limiting the ability to properly train empirical yield models across regions and years (Doraiswamy et al., 2004; Fang et al., 2011). Therefore, maintaining capabilities for satellite yield estimation that do not rely on in situ measurements for calibration are crucial for agricultural development around the world, particularly in smallholder systems (Azzari et al., 2017; Jin et al., 2019; Liu and Basso, 2020; Waldner et al., 2019).

One approach developed to efficiently and reliably estimate yields from satellite data across a range of contexts is the Scalable Crop Yield Mapper (SCYM) (Lobell et al., 2015). Rather than relying on in situ yield measurements for empirical calibration, SCYM leverages crop model simulations designed to capture key features and variation in the target crop and study area. Statistical relationships are then derived from these simulations which predict yields from the modeled crop phenology and growing season climate. The statistical models can then be applied on a pixel-by-pixel basis across regional scales to any satellite imagery able to characterize observed crop phenology, in conjunction with gridded climate datasets and crop type maps. Initial efforts demonstrated that SCYM can leverage the Landsat archive to generate multidecadal annual reconstructions of maize and soybean yields across the US Corn Belt that capture about one-third of farm yield variation (Lobell et al., 2015). SCYM has also been applied to wheat in India as well as maize and

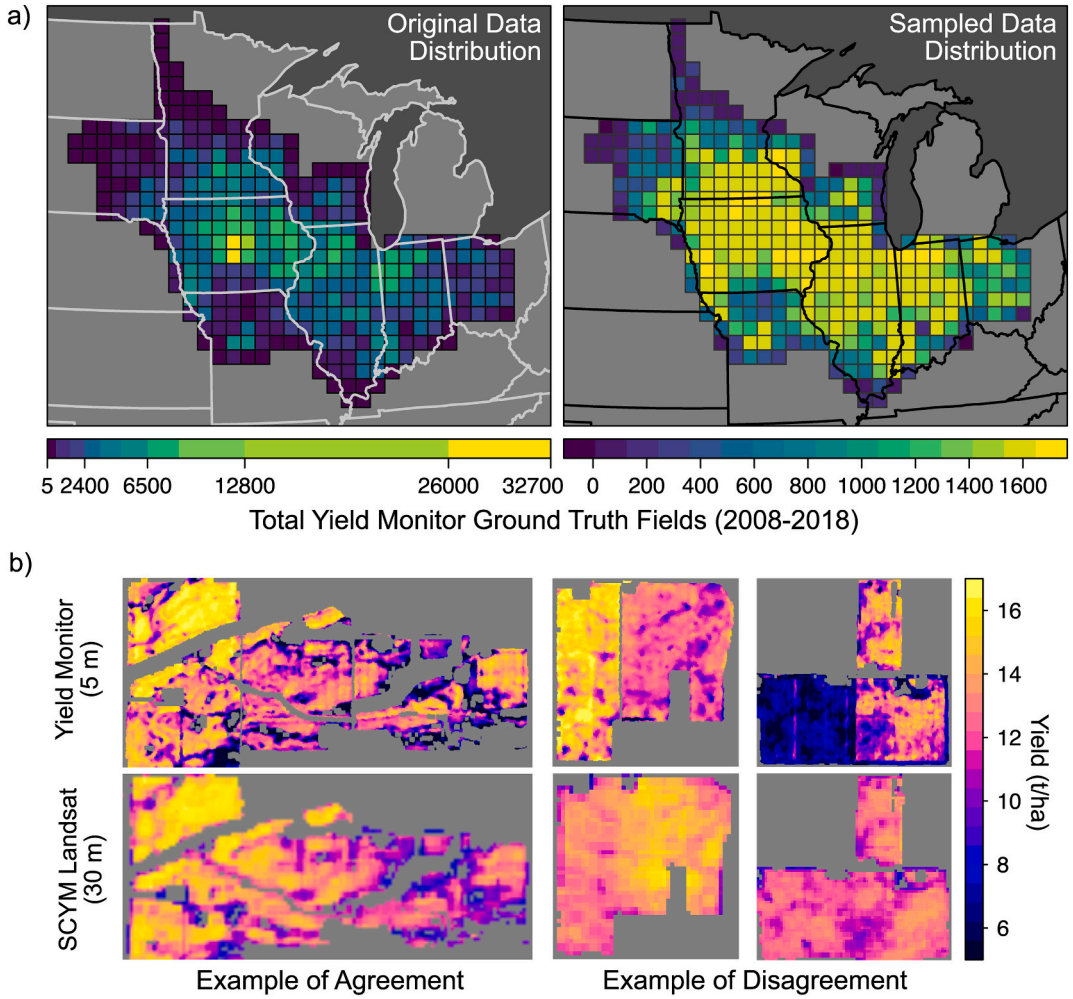


Fig. 2. Yield monitor ground truth data. a) Original data distribution and sampling scheme. Left: Density of total available yield monitor fields from 2008 to 2018 on a regular 50 km grid. Right: Density of sampled pixels, where up to 150 pixels per grid cell were randomly sampled from unique fields each year. b) Example cases of agreement and disagreement between yield monitor ground data (top row) and paired Landsat-derived satellite yield maps (bottom row) in 2018. The location of fields in panel (b) is not provided to maintain data privacy.

sorghum in sub-Saharan Africa, sometimes with methodological alterations to overcome challenges specific to each setting (Azzari et al., 2017; Burke and Lobell, 2017; Jain et al., 2016; Jin et al., 2019; Lobell et al., 2019). There remains a need to compare these various SCYM methodologies against location-specific ground data to optimize model performance for US maize applications, gain a better understanding of satellite observation requirements for yield mapping, and evaluate their utility for yield gap analyses.

Here, we take advantage of an extensive yield monitor ground truth dataset of over one million maize field observations spanning nine states in the US Corn Belt (Fig. 1) and years 2008–2018 to assess SCYM maize performance at multiple spatial scales and better understand sensitivity to factors constraining yields. We approach this in four main steps: 1) Use direct comparisons against ground data to evaluate alternative SCYM implementations, including variation in inputs such as crop phenology metrics and climate data; 2) quantify the density and timing of Landsat satellite observations needed to make robust yield predictions; 3) assess SCYM's performance relative to two alternate approaches, including a null model based on mean county yield statistics (e.g, Wolanin et al., 2020) and a random forests-based machine learning approach; and 4) evaluate SCYM's ability to capture yield responses to management, soil, and weather conditions.

For all of these questions, we focus on retrospective yield mapping with the goal to reconstruct historical yields for analysis, rather than the related problem of real-time yield forecasting that is important for food security interventions and commodity markets. Examples of analyses that use historical yield maps include targeting of sustainable intensification interventions (Jain et al., 2019) and evaluating the yield impacts of conservation agriculture techniques such as reduced tillage (Deines et al., 2019), crop rotations (Cohen et al., 2019), and cover crops (Seifert et al., 2018).

2. The Scalable Crop Yield Mapper (SCYM) in the US Corn Belt

The Corn Belt region covers approximately 1 million km^2 across the midwestern United States (Green et al., 2018). It offers an ideal setting for methods development and evaluation due to its relative richness of agricultural statistics, larger field sizes, and importance to the global supply of maize and soybeans. Typical field sizes range from ~4 to 250 ha (Yan and Roy, 2016). Therefore, moderately high resolution satellites like Landsat (30 m) are able to capture sub-field variability with minimal edge-pixel contamination. Yield heterogeneity within fields is at least half as large as overall county-level heterogeneity (Lobell and Azzari, 2017), highlighting the importance for high resolution yield maps that are accurate and validated at sub-field levels.

SCYM was first developed and applied to rainfed maize and soybeans in the three core Corn Belt states of Indiana, Illinois, and Iowa by Lobell et al. (2015). The approach relies on process-based crop simulation

models, which calculate daily crop growth based on crop physiology, temperatures, and fluxes of solar radiation, moisture, and nutrients. Specifically, SCYM uses the Agricultural Production Systems sIMulator (APSIM) model (Holzworth et al., 2014) to generate several thousand pseudo-observations of daily crop phenology parameterized to capture realistic variability in sowing dates, soil quality, and weather in the target region. Because APSIM simulates crop Leaf Area Index (LAI), a property not directly observable by satellites, LAI is translated to the Green Chlorophyll Vegetation Index (GCVI) (Gitelson et al., 2005; Gitelson et al., 2003), which is defined as:

$$GCVI = \frac{NIR}{Green} - 1 \quad (1)$$

and does not saturate as quickly as other commonly used indices such as the Normalized Difference Vegetation Index (NDVI) or Enhanced Vegetation Index (EVI) (Viña et al., 2011). Simulated LAI is converted to GCVI based on the following equation derived from empirical observations (Nguy-Robertson et al., 2012):

$$GCVI = 1.4*LAI^{1.03} + 0.93 \quad (2)$$

The simulations are then used to derive statistical relationships relating the final observed yield to GCVI-measured crop phenology and weather following the general form of:

$$Yield = \beta_o + \beta_{VI}*GCVI + \beta_w*W \quad (3)$$

where GC*VI* is a vector of crop phenology metrics and *W* is a vector of weather covariates. Both vectors can vary depending on the crop type and region in question.

To date, SCYM applications in the Corn Belt (Azzari et al., 2017; Jin et al., 2017; Lobell et al., 2015) have followed a "two-window" approach where the GCVI vector is of length two. In this formulation, the first GCVI term comes from the maximum value observed in an "early season" observation window between day-of-year (DOY) 165–200 (June 14– July 19) and the second for a "late season" observation window between DOY 201–240 (July 20 – August 28). β coefficients for the yield regression (Eq. 3) are fit for all possible combinations of DOYs from each window, thus enabling yield estimation for a flexible range of possible satellite observations, which cannot occur contemporaneously across regional scales due to variation in satellite overpass dates and cloud cover. The date and value of the maximum satellite-observed GCVI in each window is then used to select and apply the yield regression for pixel-specific observation dates.

Jin et al. (2017) improved APSIM maize simulations to better match satellite-observed crop phenology, expanded the study area to nine core states in the Corn Belt which have a high concentration of non-irrigated maize production (Fig. 1), incorporated a blending technique for Landsat- and MODIS-estimated yields (Azzari et al., 2017), and most notably, switched from directly estimating maize yields to estimating biomass, which was then converted to yield estimates with a constant harvest index. Harvest index is defined as the ratio of crop yield to aboveground biomass; this simpler approach is effective because changes in grain biomass specifically (vs total biomass) are difficult to detect in satellite data (Guan et al., 2017), and the mechanistic simulation of grain formulation in crop models tends to be less accurate than applying a harvest index to biomass, possibly due to the large number of parameters that have not been sufficiently calibrated (Jin et al., 2016). Yield maps generated with this approach captured up to 75% of variation compared to county data.

Here, we focus on the nine-state study area from Jin et al., 2017. Ground truth data (see section 3.1) overlaps the majority of corn area in these states with the exception of Michigan. We include Michigan and areas outside the ground truth domain in the final yield mapping application here to align with previous SCYM work in the region (Jin et al., 2017), although we are only able to evaluate accuracy in these regions with county yield statistics (section 3.4). Based on 2008–2018 means we extracted from the Gridmet climate dataset (Abatzoglou, 2013) for these states, growing season (June–August) mean air temperatures ranged from 14.6 to 27.3 °C. Total growing season precipitation (May – August) ranged from 152 to 728 mm.

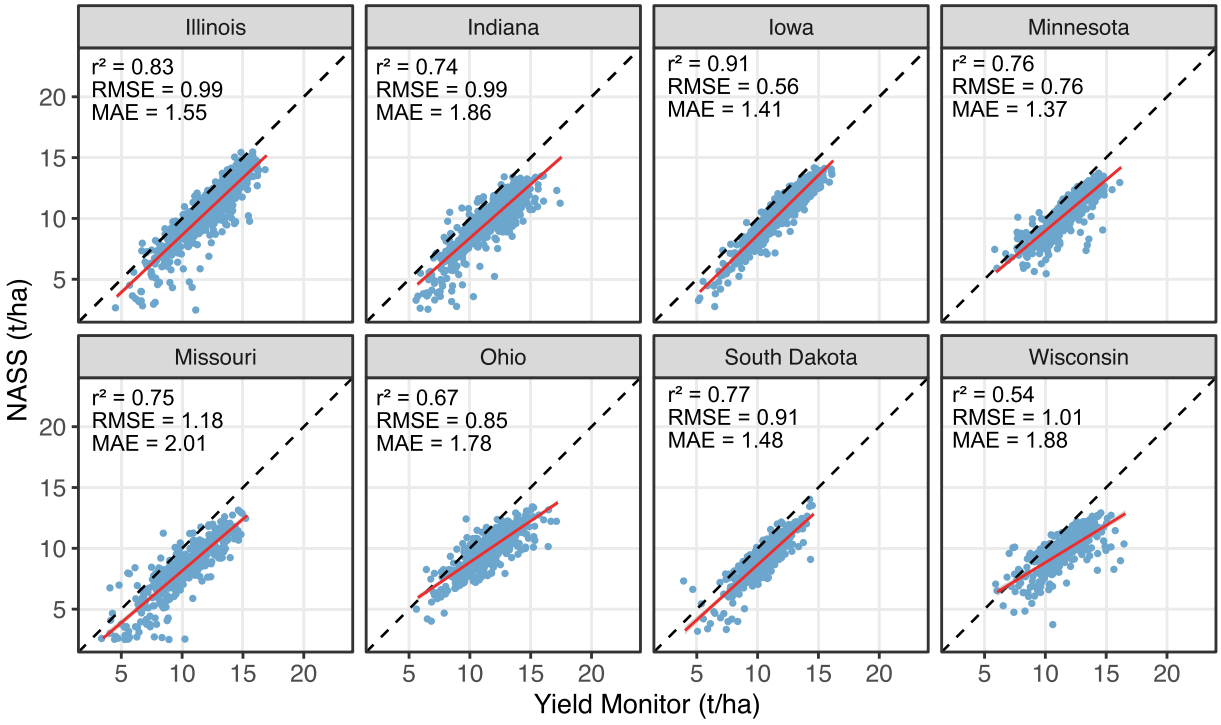


Fig. 3. County-level agreement by state between the ground yield monitor data and NASS county statistics. Agreement is for years 2008–2018; RMSE and MAE are in t/ha. Overall agreement across all county-years is $r^2 = 0.79$, RMSE = 0.95 t/ha, and MAE = 1.63 t/ha.

3. Data and methods

3.1. Yield monitor data

Through a partnership with Corteva AgriScience, we obtained yield monitor data for 1,002,958 fields from 2008 to 2018 (Fig. 2a). For each field, a yield map was available (e.g., Fig. 2b). Yield monitors do not measure grain yield directly; instead, they carry sensors that estimate harvested grain flow in weight per unit time as the combine travels across the field. This information is integrated with travel speed from the on-board GPS along with grain moisture content sensors to derive dry grain yield at a specified moisture content, which is recorded along with location every few seconds during harvest. When compared with other methods of yield assessment, yield monitors are faster, convenient to use, and provide full spatial coverage of fields at fine resolution (Fulton et al., 2018).

Raw data from yield monitors can be error prone, however, particularly if component sensors are poorly calibrated and/or from improper operation (Blackmore and Moore, 1999; Simbahan et al., 2004; Sun et al., 2013). Other factors that can affect the raw point yield estimates include sharp changes in machine speeds, a "ramping effect" that underestimates yields at the start of a harvest pass and overestimates yields at the end, and failing to adjust the swath width when the pass overlaps areas already harvested (Kharel et al., 2019; Kholsa and Flynn, 2008). In addition, failing to account for the time lag between when the grain is cut and when the grain passes the weight sensor can lead to displaced yield data (Luck et al., 2014). Nevertheless, automated processing routines that employ standardized filters to turn the raw point data into spatially continuous, gridded yield data maps have shown good performance and can improve comparability among fields and years (Kharel et al., 2019; Sun et al., 2013).

Because the large size of our yield monitor dataset prohibited field-by-field manual processing, the yield monitor data was processed in the following automated ways. Raw data from multiple harvesters were aligned to use consistent data formats and measurement units. Yield was calculated from the grain flow, width of the combine harvester, distance, and time between consecutive GPS data points. Lag time settings were adjusted by minimizing differences between points in neighboring passes, and pass overlaps were detected and resolved. Because fields were harvested at different grain moisture content levels, we calibrated yields to a standard 15.5% moisture content. We then rasterized the point data into a 5 m resolution gridded dataset, with a 5 m buffer removed from the field edges. Yield maps were then smoothed with a median filter in a 15 m moving window. Finally, we filtered anomalous values where maize yields were below 3.36 t/ha (50 bu./acre) or above 21.52 t/ha (320 bu./acre). Qualitatively, we observed fields where yield monitor yield maps and SCYM satellite yield maps showed good general agreement in yield values. There were also cases of disagreement as well as cases that appear to be remaining artefacts in the yield monitor dataset that are infeasible to manually clean in such a large dataset (Fig. 2b). Overall, when aggregated to the county level, mean yields from the monitor dataset agreed generally with the United States Department of Agriculture's National Agricultural Statistics Service (NASS) county statistics ($r^2 = 0.79$, RMSE = 0.95 t/ha) (NASS, 2019), although yield monitor yields by county were typically higher than NASS (Mean Error = 1.6 t/ha, Fig. 3). A plausible explanation for this difference is that the ground-truth yield monitor dataset may represent better performing fields than the full population of fields per county. It is unclear if these are consistently higher yielding fields or if NASS statistics provide an underestimate or incomplete picture of county level yields. In general, NASS requires survey responses from only 30 fields in a county to make a yield estimate, and the overall number of usable responses has been declining in recent years (Fig. S1).

3.1.1. Sampling approach

The full yield monitor dataset was not evenly distributed across space and time (Fig. 2a), which could bias our analyses to over-represented regions and years. Table S1 provides the distribution of field yield maps by state and year. To create a balanced data sample, we created a regular 50 km^2 grid across the study region, resulting in 353 grid cells containing yield monitor data. For each grid cell, we randomly sampled up to 150 point locations per year, with each point coming from a distinct field. For grid cells with fewer than 150 fields in a year, we randomly sampled one point from all available fields. This resulted in a more homogenous sample, although grid cells at the margins of the Corn Belt were less represented due to fewer individual fields (Fig. 2a). At each randomly sampled point, we extracted the yield value from the rasterized ground data, resampling to one 30 m pixel to match Landsat's nominal resolution.

We used this methodology to generate three pixel-based datasets for yield map evaluations. First, we created a methods validation dataset, where sample pixels were restricted to locations which had at least one valid satellite observation in each of two specified time windows during the growing season, a requirement for running previous implementations of SCYM (section 2). This allowed us to evaluate alternative SCYM approaches against a consistent dataset. The methods validation dataset included 379,387 total pixels, with annual pixel counts ranging between 30,494–37,494 (Table S2).

Second, we created a dataset specifically to evaluate SCYM accuracy in data limited regions, defined here as locations where satellite observations did not meet the "two-window" criteria described above for the methods validation dataset. This dataset was used to evaluate alternative SCYM approaches as well as identify minimum satellite observation criteria needed to make reliable yield predictions (see section 3.3). Because imagery gaps failing to meet the two-window criteria are not evenly distributed in space or time due to variable cloud cover and other satellite artefacts, we created an opportunistic sample by randomly selecting 1 pixel from all fields available in these annual imagery gaps. This generated a total sample of 146,818 pixels, with individual years ranging from 3936 pixels in 2008 to 34,865 in 2012 (Table S3), when only Landsat 7 was operational (see section 3.2.1).

Third, we created a final test accuracy dataset, again using the sampling grid to randomly select 1 pixel from up to 150 distinct fields per grid-year. This dataset was not used in method comparison and not restricted by the two-window criteria, thus providing a robust test dataset of SCYM accuracy on the preferred implementation identified through methods validation. The test accuracy dataset had 406,570 pixels, with annual pixel counts ranging between 32,826–39,921 (Table S4).

3.1.2. Field and county comparisons

To quantify the effect of spatial aggregation on accuracy metrics, we also assessed yield agreement at the field and county level. For field level metrics, we followed the grid sampling rules described in section 3.1.1 for all yield monitor fields containing greater than 20 Landsat pixels, or approximately at least 1.8 ha in area. We extracted mean field yields from the rasterized yield monitor data for up to 150 fields per grid-year, resulting in 349,770 sampled fields across years (range: 30,529-38,021 per year). For county-level comparison against the yield monitor dataset, we extracted county means for all pixels containing both SCYM and yield monitor estimates. For analysis, we included counties containing at least 330 yield monitor fields during the study period to provide on average 30 fields per county each year, resulting in 437 counties for comparison.

3.1.3. Field management sample

Field-level management data specifying seeding rate, planting date, and harvest date were obtained from Corteva AgriScience for a subset of 21,264 fields within the states of Iowa (8645 fields), Illinois (5770 fields), Indiana (4442 fields), and Minnesota (2407 fields) for the year 2018. For each field polygon, we extracted the mean yield from both SCYM and yield monitor datasets.

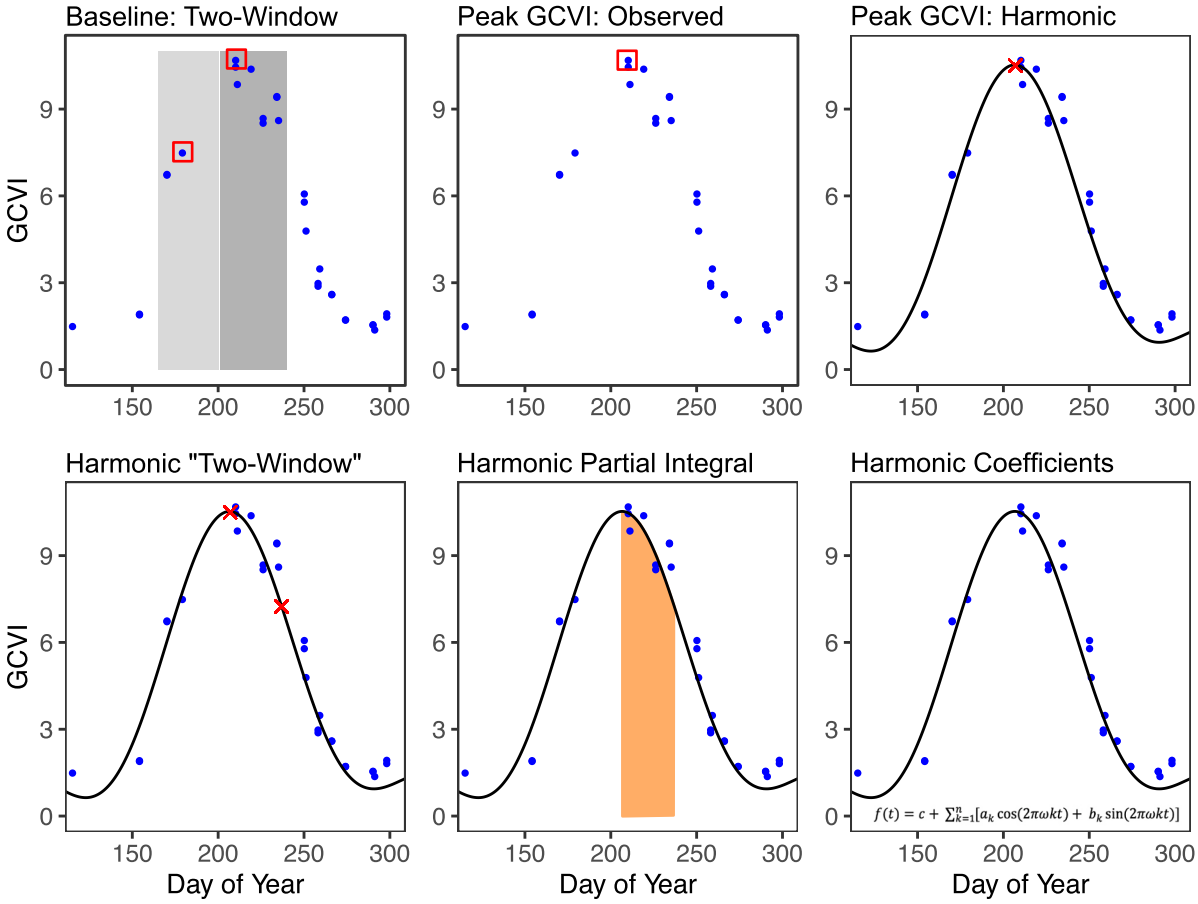


Fig. 4. Alternative crop phenology metrics evaluated in the SCYM yield mapping algorithm. Blue points indicate all clear Landsat observations for one demonstration pixel in 2017. GCVI values used in each respective SCYM implementation are highlighted in red where applicable. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.2. Evaluating alternative SCYM implementations

The spatiotemporally extensive ground truth dataset provides a unique opportunity to evaluate alternative yield mapping methods to identify a preferred approach that enables high-fidelity yield estimates across the most area. We approach this analysis in a stepwise fashion, where we start from the SCYM approach specified in Jin et al. (2017) and any beneficial changes are carried over to the next SCYM modification. SCYM performance across modifications are evaluated with r^2 (defined as the square of the correlation coefficient), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). Leading models are identified by higher r^2 and lower RMSE and MAE values when compared to the methods validation dataset. Before running accuracy metrics, we filtered out anomalous SCYM predictions, here defined as anything <0 or >25 t/ha. Anomalous predictions occurred rarely, accounting for 0.008% of pixels in the baseline model and 0.002% in the preferred model. The final selection of the preferred model is determined by performance against the data limited dataset (section 3.1.1), representative of accuracy in more challenging conditions.

We used the same set of four weather variables in all SCYM implementations, based on good performance in previous work and theoretical relationships with maize yields in the study area (Daughtry et al., 1983; Li et al., 2019b; Lobell et al., 2015): June–August rainfall, June–August solar radiation, July vapor pressure deficit, and August maximum temperature. Both June–August rainfall and July vapor pressure deficit were included as piecewise linear functions to approximate a nonlinear response, with variations above 276 mm of rainfall or below 1.6 kPa of vapor pressure deficit deemed not to affect yields. Because the study area is primarily rainfed, we do not consider supplemental water sources.

In sum, we implemented 14 alternative SCYM models: two climate sources, two harvest index values, enhanced variation in crop simulations, and six crop phenology metrics (leading to different parameterizations of the GCVI vector in Eq. 3). Fig. 4 provides an overview of the phenology metrics examined. The 14 SCYM implementations and performance are summarized in Table 1. The following subsections describe the processing of input satellite imagery and detail the alternative SCYM frameworks we tested.

3.2.1. Satellite data

Because we are particularly interested in the ability of satellites to quantify sub-field variability at scale over decadal time spans, we focus on the Landsat family of sensors; we do not consider improvements that incorporate coarse resolution but higher frequency observations such as MODIS, although we do compare the accuracy of our preferred model to the MODIS-blended product of Jin et al., 2017 (see Results section 4.2.2). We also perform a preliminary analysis to assess the performance of Sentinel-2, which has been fully operational since 2017 (see Supplemental Text S1).

We accessed all available Landsat observations through Google Earth Engine (Gorelick et al., 2017), a cloud-computing platform that provides an extensive data catalogue of publicly available satellite data along with gridded datasets of historic weather and crop types. We used all Landsat Collection 1 Tier 1 Surface Reflectance product imagery (USGS, 2017) for Landsat sensors 5 TM, 7 ETM+, and 8 OLI between January 1, 2008 – December 31, 2018 for 110 Landsat scene tiles overlying the 9-

Table 1
Comparative performance of SCYM implementations using Landsat. SCYM modifications and performance metrics across the methods validation dataset (379,387 pixels, section 3.1.1) are displayed in order of stepwise trials. ** indicates the model chosen for detailed error analysis. Table 2 describes the site locations and planting dates used, and section 3.2 provides details about the successive SCYM models.

Version	Weather (Source/# Sites)	# Planting Dates	Harvest Index	r^2	RMSE (t/ha)	MAE (t/ha)
Weather only	Gridmet, 6 sites	3	0.45	0.09	2.96	4.17
2-Window, no weather	None	3	0.45	0.14	2.93	3.23
Baseline: 2-Window	Daymet, 6 sites	3	0.45	0.31	2.63	2.23
2-Window + Gridmet	Gridmet, 6 sites	3	0.45	0.30	2.62	2.21
2-Window + Planting Dates	Gridmet, 6 sites	6	0.45	0.32	2.60	2.63
2-Window + Additional site	Gridmet, 7 sites	3	0.45	0.34	2.57	2.09
2-Window + Apsim Variation	Gridmet, 7 sites	6	0.45	0.36	2.53	2.40
2-Window: Harvest Index	Gridmet, 7 sites	6	0.50	0.36	2.53	2.13
Peak GCVI: Observed	Gridmet, 7 sites	6	0.50	0.35	2.55	2.25
Peak GCVI: Harmonic	Gridmet, 7 sites	6	0.50	0.39	2.47	2.00
Harmonic Coefficients	Gridmet, 7 sites	6	0.50	0.39	2.47	2.13
Harmonic 2-Window	Gridmet, 7 sites	6	0.50	0.40	2.44	2.04
Harmonic Partial Integral**	Gridmet, 7 sites	6	0.50	0.40	2.45	1.96
Harmonic Partial Integral, no weather	None	6	0.50	0.39	2.47	2.08

state study area. These products have been corrected for terrain, radiometric, and atmospheric artefacts, and they have a resolution of 30 m. Despite several differences in sensor specification, vegetation indices derived from the three sensors are comparable without modification for similar applications (Vogelmann et al., 2015). For each scene, we applied the accompanying cloud mask from the "pixel_qa" band to keep only "clear" pixels. For each location in our point sample, we then extracted the full time series of valid observations for each year at 30 m resolution and calculated the GCVI for each date.

All Landsat satellites follow a 16-day revisit cycle, which is offset by eight days when two satellites are operating simultaneously. This nominal 8-day revisit period applied for all years in our study except 2012, when only Landsat 7 was operational. The revisit period can be shortened through path overlap among adjacent satellite tracks and lengthened by cloud cover or the failure of the scan-line corrector in Landsat 7, which results in data gaps as you move away from the center of each scene tile. Overall, the number of clear annual observations per sampled pixel ranged from 2 to 50, with a median of 20.

3.2.2. Baseline model and alternative climate data sources

We first established baseline accuracy using the two-window SCYM implementation in Jin et al. (2017) applied to Landsat data. Here, the SCYM statistical models were derived from 2100 APSIM simulations. Fifty simulations were run per year for 2007–2013 at each of six weather sites, with variation randomly drawn from three sowing dates, three cultivars, and a range of fertilizer rates, sowing densities, and soil conditions at sowing (Table 2). We applied the baseline model using both Daymet ~1 km gridded climate reanalysis data (Thornton et al., 2016) and the ~4 km Gridmet product (Abatzoglou, 2013), since recent evidence suggests that Daymet can have meaningful biases in solar radiation (Slater, 2016). When compared to station data from 24 sites in Iowa across 2002–2018, we found Gridmet had higher agreement in solar radiation and daily precipitation (Fig. S4).

3.2.3. Enhanced variation in crop simulations and harvest index adjustment

To perform well, the crop simulations need to encompass likely variation in the study area. The three sowing dates used in baseline simulations represent the 10th, 50th, and 90th percentiles in Indiana, Iowa, and Illinois (Lobell et al., 2015) but fail to capture the full variation present within the study area (NASS, 2019; Fig. S5) We therefore tested a SCYM implementation with three additional planting dates in APSIM simulations (Table 2), increasing the number of simulations per site-year from 50 to 100. Similarly, to better represent weather variation in the expanded nine-state study region, we added an additional northern weather site to the original six sites, bringing the total number of APSIM simulations to 4900 (Table 2).

The harvest index quantifies the grain proportion of aboveground biomass. In the baseline model, it is set to 0.45 following Jin et al. (2017). Other evidence suggests that the harvest index reached ~0.50 in the early 2000s and has been relatively stable since then (Mueller et al., 2019). We therefore test an implementation of SCYM with a uniform harvest index increased to 0.50. Although the harvest index can vary across space (Zhu et al., 2019), we did not seek to tune the harvest index, reasoning that SCYM applications in data-limited regions would lack the information necessary to make specific adjustments.

Table 2
Summary of APSIM crop simulations. The Baseline column describes the original set of 2100 simulations used in Jin et al. (2017). The Additions column described additional simulations intended to enhance variation for training the SCYM yield model. Sections 3.2.2–3.2.6 provide details about successive SCYM models.

Factor	Baseline	Additions	Comments
Years	2007–2013		
Weather sites	Newton, IA (−93.1 E, 41.7 N) Marshalltown, IA (−92.9 E, 42.1) Clinton, IL (−89.0 E, 40.1 N) Chenoa, IL (−88.7 E, 40.7 N) Marion, IL (−85.7 E, 40.6 N) Munice, IN (−85.3 E, 40.2 N)	Benson, MN (−95.6 E, 45.3 W)	
Fertilizer rates	200, 300		kg of urea N per ha
Sowing dates	Apr-24, May-1, May-15	Apr-15, May-25, Jun-5	
Sowing density	4.5, 6, 7.5, 9		plants per m^2
Cultivar choice	Pioneer_short, Pioneer_3394, Pioneer_long		Refer to Jin et al., 2017 for details
Soil water at sowing	60, 80, 100		% of total water holding capacity
Total simulations	2100 (50 per site-year)	2600	Total simulations with additions: 4700 (100 per site-year)

3.2.4. Annual peak greenness models

The baseline model uses a two-window approach (section 2). However, when applying the SCYM approach to smallholder systems in Africa, more frequent cloud cover and variable planting dates made the two-window approach less feasible; instead, Jin et al. (2019) used only one term in the GCVI vector in the yield regression (Eq. 3), which represented the peak value derived from a harmonic regression fit to all available imagery. This "peak VI" approach has been used in other yield estimation applications, although a comparison for wheat prediction in Australia using crop model simulations found it performed slightly worse than the two-window approach (Waldner et al., 2019).

Here, we tested two "peak GCVI" approaches that allowed us to use a single SCYM yield regression for each version, in contrast to the day-of-observation specific regressions applied in the baseline two-window approach. The first one was data-driven, where we extracted the maximum observed GCVI value from the available annual satellite time series. For the second approach, we derived the peak GCVI value from a harmonic regression fit to all available annual imagery. Specifically, we used a discrete Fourier transform to summarize the seasonal phenology of GCVI in the form:

$$f(t) = c + \sum_{k=1}^{n} [a_k cos(2\pi\omega kt) + b_k sin(2\pi\omega kt)] \quad (4)$$

Where $f(t)$ is the time-dependent value of GCVI, t is date of observation expressed as a fraction of the calendar year between 0 and 1, n is the number of harmonic terms, ω is the frequency, and a_k, b_k, and c are the cosine, sine, and intercept coefficients, respectively. We set n to 2 and ω to 1.5, which were found to minimize the mean-squared error without overfitting for crops the US Corn Belt (Wang et al., 2019). We then used the resulting regression with fitted coefficients to interpolate a daily GCVI value for each pixel and extracted the maximum value between days-of-year 122 and 274 (May 1 – September 30). To maximize the correspondence between the SCYM yield regression and satellite data processing, we also fit the harmonic regression to each APSIM simulation and used the fitted peaks to derive Eq. 3.

3.2.5. Harmonic metrics models

Yield estimation methods focusing solely on the peak VI value can underperform because they omit information from later grain filling stages (Waldner et al., 2019). Similarly, smoothing the time series of satellite data is helpful to address inconsistency in observation frequency and timing due to clouds and other sensor artefacts (Skakun et al., 2019).Therefore, we also used harmonic regression to test SCYM models that used more informative metrics with consistent timing across pixels. We tested three implementations for the GCVI vector of Eq. 3 (Fig. 4): 1) a modified two-window approach with two GCVI terms from the peak fitted GCVI and the fitted GCVI 30 days following; 2) an adapted "partial integral" approach where we found the area under the curve between the fitted peak and 30 days after, thus providing one GCVI term for Eq. 3 but capturing more information about the shape of the phenology curve; and 3) a harmonic coefficient approach, where we used the five harmonic coefficients fit in Eq. 4 directly in the GCVI vector.

3.2.6. Influence of weather covariates

Recent work on Australian wheat yields suggests as much as half of the accuracy in a two-window approach may be attributed to weather covariates, while weather contributed little to accuracy for more advanced phenology metrics (Waldner et al., 2019). To better evaluate the accuracy contributions from the GCVI and weather vectors, we included a model with only the weather vector (no GCVI terms) and two models with only the GCVI vector, one based on the two-window approach and one based on a partial integral derived from the harmonic regression.

Table 3
Summary of six alternative satellite observation window criteria. Median and interquartile range (IQR) values were derived from the day of observed peak GCVI from all Landsat observations for the methods validation sample between 2008 and 2018.

Name	Day-of-Year Range	# Days	Source
Median +/− 10	195–215	21	Median date of maximum GCVI, Landsat 2008–2018
IQR	194–217	24	IQR for date of maximum GCVI, Landsat 2008–2018
IQR +/− 5	189–222	34	IQR for date of maximum GCVI, Landsat 2008–2018
IQR +/− 10	184–227	44	IQR for date of maximum GCVI, Landsat 2008–2018
Wide window	180–250	71	Useful yield prediction window, Gao et al., 2018
Unrestricted	1–365	365	No restrictions

3.3. Minimum satellite observation requirements

For accurate yield estimation, the density of satellite observations needs to be sufficient to capture the evolution of the GCVI curve over time, particularly during the peak growing season. The sufficient number, however, is not well quantified. We approached this question in two steps. First, we took three of the top-performing models against the methods validation dataset and evaluated their accuracy against the data limited dataset described in section 3.1.1. We then identified the preferred SCYM implementation as the best performing model against this dataset. Second, we took this preferred SCYM implementation and conducted an accuracy analysis along a range of minimum satellite observation criteria (described below), where we considered both overall accuracy and the amount of pixels failing to meet these criteria.

The preferred model relies on harmonic regression (see Results). For fitting harmonic regressions, one must have at least as many observations as the numbers of parameters to fit; in our second order harmonic equation (Eq. 4), this number is five. However, this minimum threshold can still fail to capture crop phenology, particularly if poorly timed. Recent work suggests that at least 15–20 observations are needed to fit a 5-parameter non-linear harmonic regression and adequately capture the magnitude and timing of peak growth of crops in our region, and that even with this density of observations, a 5-parameter linear harmonic regression (used in this paper) underperforms at this task (Roy and Yan, 2018). Such requirements would limit the ability of Landsat alone to estimate yields, as also suggested in other work (Gao et al., 2018). Roy and Yan (2018), however, focused on the Normalized Difference Vegetation Index (NDVI), which is known to saturate in crop canopies, and evaluated only the ability to describe the peak, not the ability to estimate yield outcomes.

To better define the observational requirements for accurate maize yield estimation, we leveraged our large ground dataset to evaluate accuracy and coverage area as a function of the number and distribution of observations. We defined two criteria: the minimum number of observations per year, ranging from five to 20+; and the presence of at least one clear observation during six alternative definitions of a "key crop window" (Table 3). We designed these six alternative periods to range from non-restrictive (no window) to more restrictive and based them upon the observed dates of peak GCVI values from Landsat 2008–2018 time series for the methods validation dataset as well as literature information. We then fully crossed these criteria to create a sensitivity analysis with 96 alternative requirements. We filtered both the gap accuracy and methods validation datasets to determine the proportion of pixels meeting each requirement and the associated yield estimation accuracy for those pixels. Finally, we visually identified the criteria that provided the most accuracy with the least cost to coverage by plotting accuracy vs. percent of pixels unable to meet the requirement.

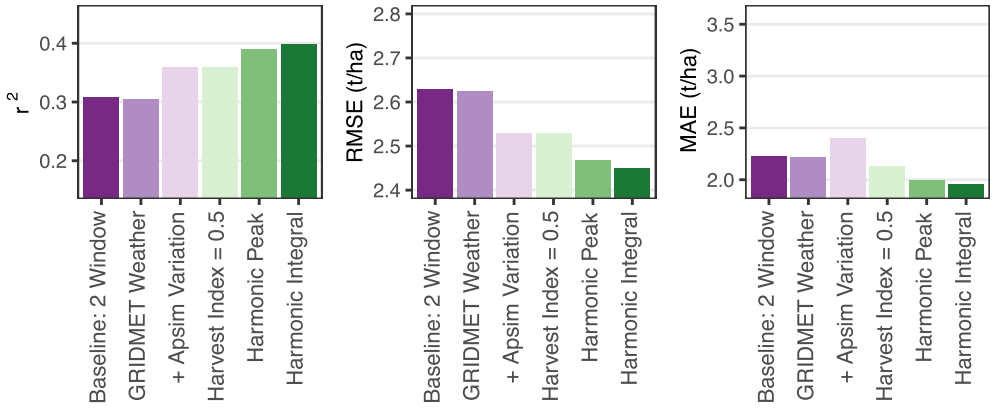


Fig. 5. SCYM performance improvements Landsat accuracy statistics for a subset of SCYM implementations (see Table 1) derived from the methods validation dataset (n = 379,387 pixels) from 2008 to 2018.

3.4. Regional yield maps and final accuracy evaluation

We applied the final preferred SCYM implementation to all Landsat imagery for pixels classified as maize by the USDA's cropland data layer (CDL) to create annual yield maps at 30 m resolution from 2008 to 2018 across the 9-state study area. We applied the filtering criteria identified in section 3.3 to exclude any pixels lacking sufficient satellite data for reliable yield estimation. To quantify how restrictive this was compared to the previous two-window approach, where any pixel lacking an observation in either window was unable to be used, we found the annual maize area each year unable to generate a yield estimate. We evaluated the final accuracy of the preferred SCYM approach using the pixel test dataset (section 3.1.1). We also compared our annual yield maps at the field and county levels to assess the scale dependency of accuracy metrics. For county level comparisons against the yield monitor data, we only considered SCYM estimates overlying locations with yield monitor data. We also assessed county accuracy against NASS county yield statistics by taking the mean of all SCYM estimates in each county.

Finally, we compared our scalable yield mapping approach against both a null model and a ground-calibrated machine learning model to better understand SCYM's relative performance. For the null model, we used NASS county statistics to predict the annual yield for each county based on that county's mean across all other years in the study period. We compared the performance of this null model against SCYM yield predictions at the pixel, field, and county scales; for pixel and field scales, each observation was given the null model county prediction.

We also developed a ground-calibrated machine learning model to better understand the data requirements needed to achieve a similar performance to SCYM and explore the upper accuracy limits one might expect in this region based on our ground data. For this comparison, we chose to use a random forests regressor (Breiman, 2001). We did not explore alternate machine learning approaches since random forests are straightforward to implement and have outperformed other machine learning and multiple regression approaches in similar contexts (Kayad et al., 2019). We also used the same GCVI and weather features from the best performing SCYM implementation rather than exploring alternate sets of features. Our goal was to understand how the training sample size and their spatiotemporal distribution affected model performance. We used the randomForest package (Liaw et al., 2002) in R (R Core Team, 2014) to train a suite of random forests regressors, all of which were parameterized with 200 trees, bag fraction of 0.63, minimum leaf size of 5, and 2 variables per split. We implemented a two-stage sensitivity analysis. First, we varied the number of samples randomly drawn from the methods validation dataset (section 3.1.1) to create nine models. Sample sizes used were 50, 100, 500, 1000, 10,000, 50,000, 100,000, 150,000, and 200,000. For the sample size that achieved accuracy most similar to SCYM, we then performed a secondary analysis by restricting that number to single years or to single states. This allowed us to test how well a model trained on data from one year, or a smaller geographic region, generalized to the full study area and time period. All model accuracies were evaluated against the test dataset spanning the full study region and time period.

3.5. Yield responses to management and environmental factors

We compared the yield responses of the preferred SCYM approach and ground data to underlying variation in management and environmental conditions as both a secondary accuracy assessment and to better assess SCYM's ability to identify yield gaps and factors limiting yields. Similar to many other sources of ground truth yield data, yield monitor data suffers from a number of factors that contribute to noise and uncertainty in its accuracy (section 3.1). A secondary accuracy metric for satellite products, therefore, is to compare how each dataset responds to other variables known to influence yields (Burke and Lobell, 2017).

Here, we compared yield responses to two sources of variation. First, we used the 2018 field management data (section 3.1.3) to compare how field-level yield means respond to variation in seeding rate, planting date, and harvest date. Second, we used the pixel test dataset to compare responses to environmental variation from 2008 to 2018. Specifically, we examined responses to three soil and three weather properties. For each pixel, we derived the soil available water capacity in the top 1 m of soil and the soil productivity index for maize from the gSSURGO soil database (NRCS, 2016). We obtained May mean soil moisture and July mean water deficit from the Terraclimate gridded climate dataset (Abatzoglou et al., 2018), and total growing season precipitation (May – August) and mean June–August temperature from the Gridmet dataset.

Notably, none of these variables are directly included in the SCYM yield regression, and SCYM has no information about on-the-ground field-specific management nor soil properties. We used the R package "mgcv" (Wood, 2017) to implement a General Additive Model (GAM) with cubic splines and a basis dimension of five to fit a yield-variable relationship for each property to allow for non-linear effects.

Methods comparisons, accuracy evaluations, and figures were conducted in R (R Core Team, 2014) using the tidyverse suite of packages (Wickham et al., 2019) and spatial packages sf (Pebesma, 2018), rgdal (Bivand et al., 2014), and raster (Hijmans, 2014) on satellite time series exported from Google Earth Engine at each location described in our sampling approach (section 3.1.1); final SCYM map products were produced in Google Earth Engine. Yield map products are available publicly at an aggregated 10 km spatial resolution for privacy concerns (Zipper et al., 2019) as described in the Data Availability section.

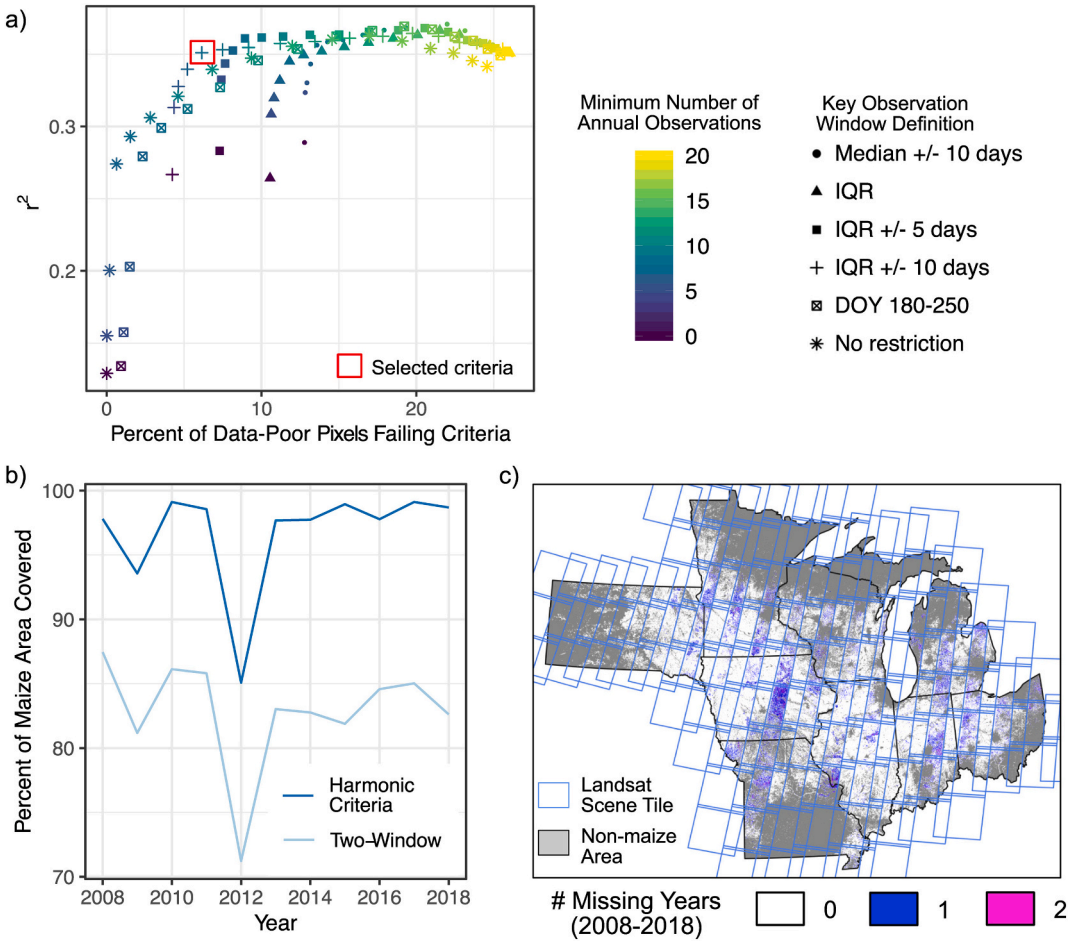


Fig. 6. SCYM coverage in the study area. a) Identification of minimum satellite observation criteria based on the data limited sample (section 3.1.1). Window criteria are defined in Table 3. b) Annual maize area (%) that met the satellite criteria to enable SCYM yield estimation. c) Spatial distribution and annual frequency of areas that failed to meet the satellite observation criteria; Locations failing to meet criteria most often occurred outside of Landsat scene overlap areas.

4. Results and discussion

4.1. Accuracy assessment across alternative SCYM yield mapping approaches

Using only Landsat imagery from 2008 to 2018, the two-window baseline SCYM model explained 31% of yield variation at the 30 m pixel level across the methods validation dataset ($r^2 = 0.31$, RMSE = 2.63 t/ha, MAE = 2.23 t/ha). Through our evaluation of alternative approaches, we identified several better performing models able to capture ~40% of pixel-level yield variation (Table 1). The largest accuracy increases came from enhancing the variation of APSIM crop simulations used to derive the SCYM yield regression and from using harmonic regressions to better characterize the annual time series of satellite observations (Table 1, Fig. 5). We also found that increasing the harvest index to 0.5 reduced MAE, and that the Gridmet gridded weather dataset performed similar to Daymet overall but did improve accuracy for the 2017–2018 year subset examined in the preliminary Sentinel-2 analysis (Fig. S3).

The five SCYM models based on harmonic regression performed similarly well against the methods validation dataset (Table 1), which was limited to locations where the baseline two-window approach could be implemented. In these relatively "data rich" locations, not including the weather vector in the harmonic partial integral implementation performed only slightly worse than the corresponding model retaining weather information ($r^2 = 0.39$ vs $r^2 = 0.40$, respectively, Table 1). In contrast, omitting the weather variables from the baseline two-window approach performed poorly ($r^2 = 0.14$), but better than a model reliant only on weather ($r^2 = 0.09$). This is consistent with recent findings that more advanced phenology metrics can perform well without additional weather variables, but performance for simpler metrics such as the two-window approach is approximately halved when weather information is omitted (Waldner et al., 2019).

4.1.1. Data limited locations and preferred SCYM model

We selected three best-performing harmonic models for evaluation against the data limited dataset: the two-window harmonic model, partial integral model with weather, and partial integral model without weather. As anticipated, overall performance on this dataset was lower than on the methods validation dataset. The partial integral model and harmonic two-window model performed similarly ($r^2 = 0.33$). The partial integral model without weather, however, performed worse ($r^2 = 0.28$). This suggests that the inclusion of the weather vector in the SCYM regression is particularly helpful when there are fewer satellite observations available.

Due to its slightly better performance against the methods validation dataset and better robustness in data limited locations, we concluded that the harmonic partial integral approach including the weather vector is the preferred SCYM yield model.

4.1.2. Minimum satellite requirements

We used this preferred SCYM model to identify minimum satellite imagery requirements for effective yield estimates. When compared against the data limited sample, we found that locations with at least

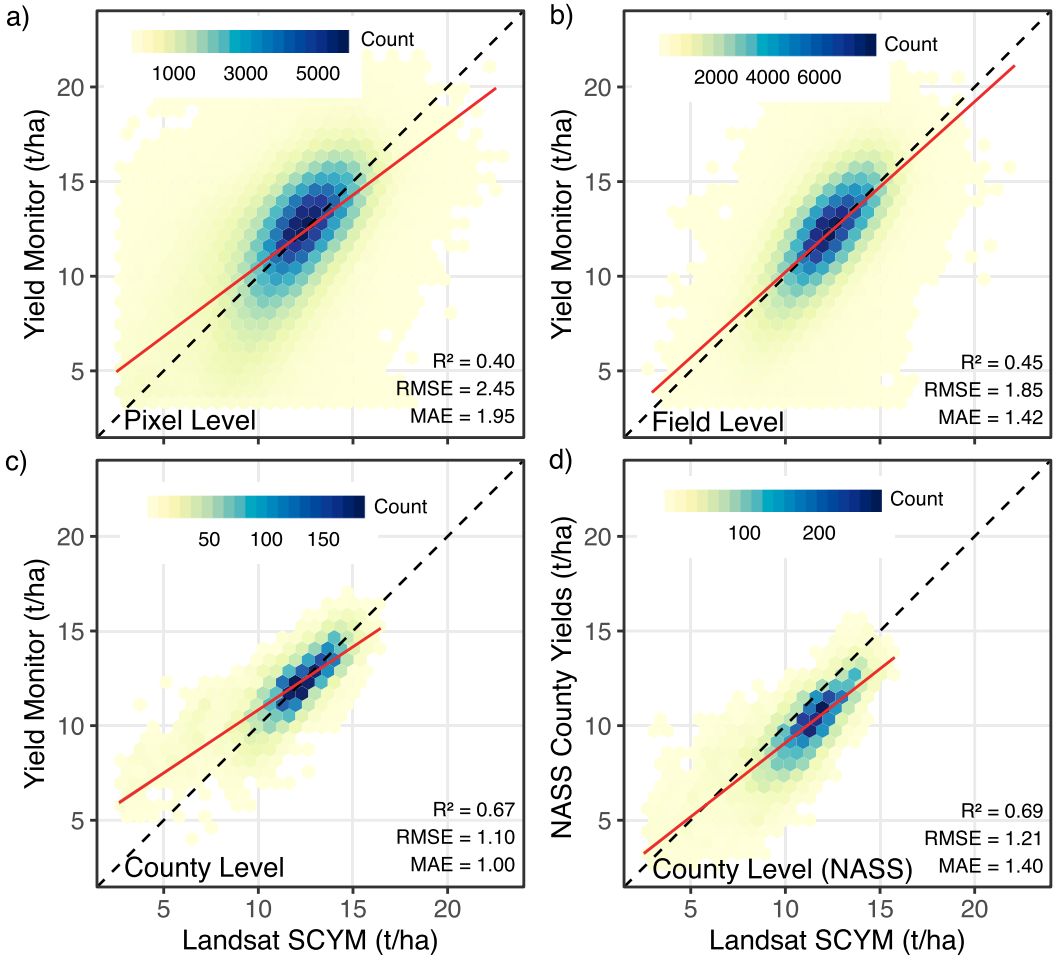


Fig. 7. Satellite yield mapping accuracy with Landsat at multiple scales, 2008–2018. a) Pixel-level accuracy metrics from the yield monitor test accuracy dataset. b) Field-level accuracy metrics derived by aggregating yield monitor and SCYM yield maps to field means. c) County-level accuracy derived by aggregating yield monitor and SCYM yield maps to county means for locations with yield monitor data (437 counties). d) County-level accuracy derived from USDA NASS county statistics compared to SCYM mean yields by county (761 counties). All panels: RMSE and MAE are provided in t/ha.

eight annual observations and at least one observation within a 44 day critical window (DOY 184–227, or July 3 – August 15) generated the highest accuracy (r^2 = 0.35) while excluding the fewest pixels (6.1%, Fig. 6a). Because more stringent requirements produced only marginal gains in accuracy but excluded as many as 25% of pixels, we concluded that these criteria produced the optimal results by increasing confidence in yield estimates and maximizing coverage in application.

This analysis indicates that credible satellite yield estimations can be made with more sparse time series than previously suggested. Roy and Yan (2018) found that 15–20 annual observations are needed to describe the phenology peak with harmonic regression, although they were not able to assess requirements for yield estimation specifically. It is possible we were able to reduce the absolute number needed by combining the annual minimum threshold with the critical window definition. Gao et al. (2018) conclude that Landsat alone would be insufficient for maize yield estimation in Iowa, but our Landsat-based SCYM approach achieved similar county-level accuracies as their Landsat-MODIS fusion approach (see section 4.2.2), and they were unable to evaluate field or pixel level accuracy. Waldner et al. (2019) recommend that harmonic or integral-based metrics be used only with clear imagery available at least every five days for wheat yield estimation in Australia, but we show these can perform well at scale with less dense imagery, albeit for maize.

For applications which do not extend back in time past 2017, the Sentinel-2 constellation provides a promising imagery source. Sentinel-2 provides nominal 5-day imagery at 10–20 m resolution (ESA, 2020). In a preliminary analysis relying on informal atmospheric correction, we found that SCYM applied to Sentinel data performed similarly to Landsat in 2017–2018, but the density of observations was able to cover more area consistently (see Supplementary Text S1; Fig. S3). In other contexts, datasets which harmonize Landsat and Sentinel data can outperform single-sensor applications (Skakun et al., 2019). These harmonized datasets offer promising opportunities for yield mapping in recent and future years.

4.2. Assessment of the preferred SCYM yield mapping approach

4.2.1. Accuracy across spatial scales

The final SCYM yield maps were produced with the harmonic partial integral method applied to all available Landsat imagery (Fig. 1). The SCYM yield regression was derived from the fully expanded set of crop simulations, and we used the Gridmet climate dataset and a harvest index = 0.5. We applied SCYM to all maize pixels that had at least eight clear satellite observations during the calendar year and at least one observation within the critical window of DOY 184–227, or July 3 – August 15. When assessed with the test accuracy dataset, SCYM was able to explain 40% of pixel-level yield variation across the 9-state region between 2008 and 2018 (RMSE = 2.45 t/ha, MAE = 1.95 t/ha, Fig. 7a). Fig. S6 provides a breakdown of pixel-level accuracy by state and year.

To understand factors contributing to yield variability not detected well by SCYM, we compared outlier pixels (defined as pixels where the

difference between SCYM and ground truth was outside the 1st and 99th percentiles, or approximately >6 t/ha) to pixels where prediction differences were within the interquartile range (−1.65 to 1.41 t/ha; Fig. S7). Although we failed to identify a consistent set of conditions that led to poor agreement with ground truth, SCYM outlier overestimates were skewed toward poorer quality soils, while outlier underestimates skewed toward high quality soils. SCYM also tended to overestimate yields at higher latitudes, which suggests that adjusting the harvest index in a scalable way may be promising for future work. We also found that SCYM outlier underestimations tended to have fewer cloud-free Landsat observations and occurred in low rainfall conditions. However, 47% of outlier underestimations occurred in 2012, when an unusually strong drought coincided with only one active Landsat satellite, resulting in fewer observations (see 3.2.1). It is plausible that SCYM may have overly penalized low rainfall and high vapor pressure deficit conditions, in part because it was naïve to supplemental water sources such as shallow groundwater or irrigation. How this interacted with reduced satellite observations is unclear. Overall, the wide variability in the point cloud for the pixel-level assessment (Fig. 7a) along with anecdotal examples of mismatches between SCYM satellite estimates and ground data (e.g., Fig. 2b) suggest that while satellite yield maps are robust for inferences in "big data" contexts when aggregating performance over large field samples, applications for location-specific analyses such as precision agriculture recommendations for specific field areas should be pursued with caution.

With the new minimum satellite observation criteria, we were able to estimate yields on average for 96.6% of annual maize area (range 85.1% - 99.1%) compared to 82.4% for the previous two-window approach (Fig. 6b). Minimum coverage occurred in 2012, when only Landsat 7 was operational, thus reducing the number of available satellite observations. In general, pixels failing to meet satellite observation requirements were located in the non-overlapping regions of Landsat scenes, which have a lower potential number of cloud-free observations, and most of these pixels failed in only one year (Fig. 6c).

Spatial aggregation increased accuracy metrics, particularly when aggregated to political boundaries (Fig. 7). Assessment at the field level increased r^2 from 0.40 to 0.45 and resulted in a corresponding reduction in RMSE (2.45 to 1.86 t/ha) and MAE (1.95 to 1.43 t/ha). Field-level yield agreement was less dispersed than at the pixel-level but still displayed a large spread around the 1:1 line, likely due to data artefacts from both yield monitor and satellite sources. When aggregated to the 437 counties in which we had sufficient yield monitor data, SCYM captured 67% of yield variation with corresponding reductions in spread (RMSE = 1.10 t/ha, MAE = 1.00 t/ha).

When comparing 761 counties with available NASS county yield statistics in the study region, SCYM explained 69% of yield variation across all county-years (RMSE = 1.21 t/ha, MAE = 1.40 t/ha). The overall relationship to NASS data was stronger than to yield monitor data at the county level, even though the preferred SCYM specification was selected based on yield monitor validation data. We did, however, see a slight overestimation of yields compared to county level statistics which is reflected in a higher MAE (Fig. 7d). This may be a result of the higher yields observed in ground-truth yield monitor fields compared with NASS county numbers (section 3.1, Fig. 3). It's unclear if the bias results from selecting our model based on this data, or if NASS statistics provide an incomplete picture of county yields due to the declining survey response rate (Fig. S1) and low minimum requirement of 30 fields per county.

Overall, the comparison of accuracy by spatial aggregation gives an unsurprising but important result: when high-resolution yield maps are aggregated to political boundaries for accuracy assessment, the resulting accuracy cannot be inferred to apply to smaller field-level or pixel-level scales. We believe that agreement at aggregated scales is still informative and useful when it is the best available comparison, but researchers should strive to provide an assessment of yield mapping methods at a comparable scale for high resolution products.

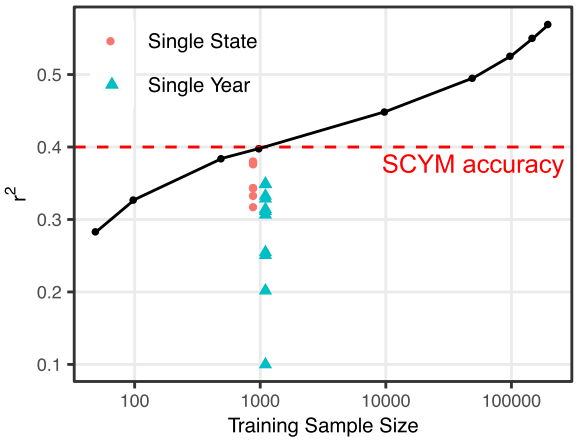


Fig. 8. Sensitivity analysis of random forests pixel-based yield models by training sample size and distribution. The black line shows accuracy by training sample size for pixels randomly sampled from the full methods validation dataset. Note the x-axis is logarithmic. Pink points show accuracy for models trained on 1000 pixels sampled from only one state, and teal points show accuracy for models trained on 1000 pixels sampled from only one year. Model accuracy was evaluated with the same test dataset used for final SCYM evaluation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.2.2. *Comparison with other yield mapping approaches*

The updated SCYM approach using only Landsat data compares favorably with past SCYM efforts, our alternative null and ground-calibrated models (section 3.4), and other work in the study area incorporating additional satellite imagery such as MODIS. When compared with previous SCYM yield mapping approaches, we found that the preferred harmonic approach provides higher accuracy and increased areal coverage. The previously published version of SCYM, which blends Landsat and MODIS yield predictions (Jin et al., 2017), does reasonably well against the methods validation sample ($r^2 = 0.34$, RMSE = 2.53 t/ha, MAE = 2.13 t/ha). Due to the relatively strict requirement of having one observation in each of two separate windows, however, the previous method covered 14.2% less area on average for 2008–2018 (Fig. 6b). The previous version does outperform the preferred harmonic approach presented here when compared against NASS county statistics for the same time period and region ($r^2 = 0.74$, RMSE = 1.11 t/ha, MAE = 0.876 t/ha). The MODIS predictions (blended with Landsat predictions to produce final predictions in Jin et al., 2017) on their own did poorly when compared with our higher resolution 30 m pixel test dataset ($r^2 = 0.25$, RMSE = 2.72 t/ha, MAE = 2.49 t/ha).

Our null model based on county yield statistics performed poorly across all scales when compared to the preferred SCYM approach. Null model agreement with annual NASS statistics was best at the county level with $r^2 = 0.35$ (RMSE = 1.74 t/ha, MAE = 1.29 t/ha), capturing approximately half as much variation as SCYM. Performance was even worse when compared with yield monitor data aggregated to counties ($r^2 = 0.13$, RMSE = 1.77 t/ha, MAE = 2.02 t/ha). This supports the idea that annual greenness metrics from satellites improve accuracy when generating county yield models (Li et al., 2019b; Wolanin et al., 2020). The null model performed poorly against yield monitor data at both the field ($r^2 = 0.08$, RMSE = 2.39 t/ha, MAE = 2.41 t/ha) and pixel ($r^2 = 0.04$, RMSE = 3.08 t/ha, MAE = 2.76 t/ha) scales. This was expected since county statistics lack granularity. In contrast, satellite-derived yield maps such as SCYM enable yield heterogeneity analyses both between and within fields, including fine-scale factors such as soil properties or management practices.

The ground-calibrated random forests model outperformed SCYM when trained on data from 200,000 fields ($r^2 = 0.57$, RMSE = 2.07 t/ha, MAE = 1.55 t/ha; Fig. 8). We consider this an estimate of the lower

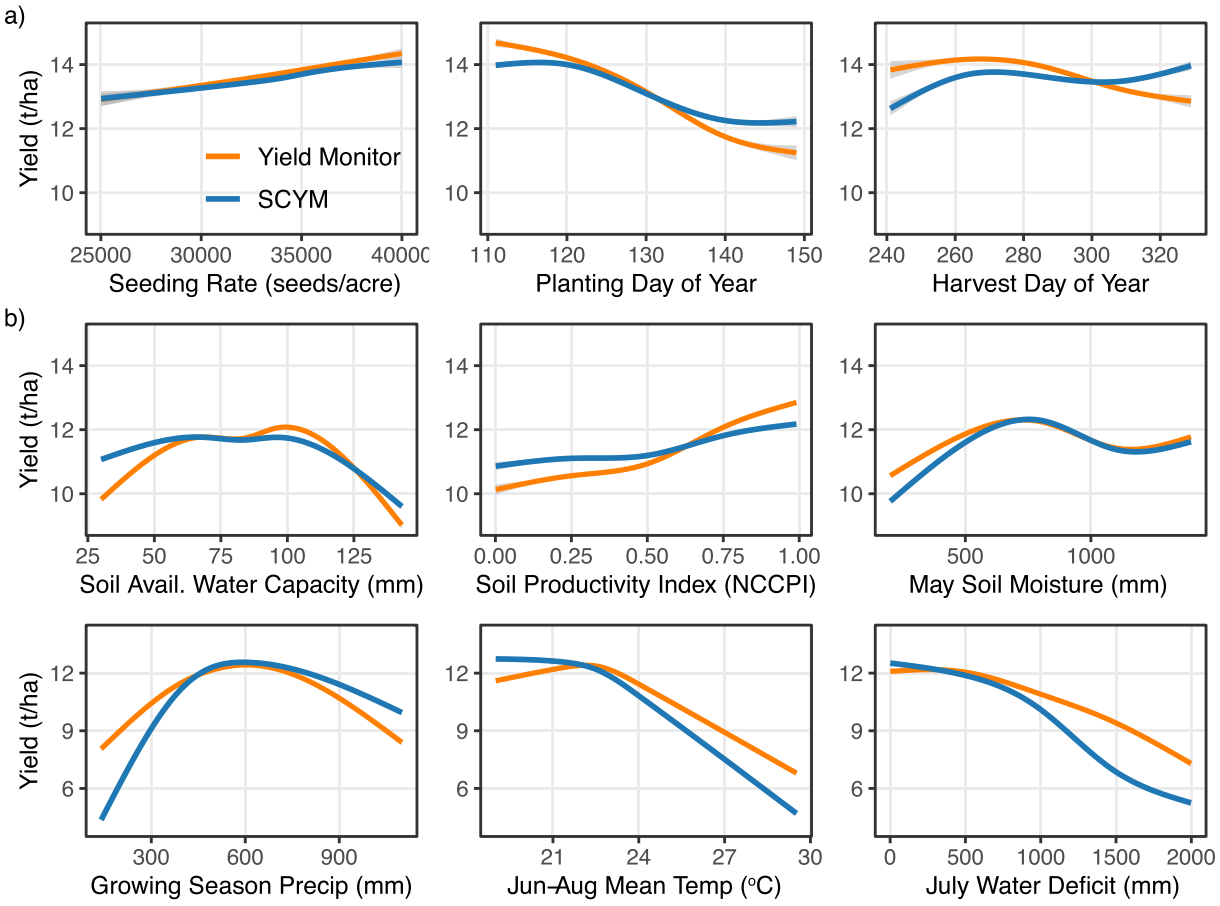


Fig. 9. Yield responses to variation in management and environmental variables by dataset. Yield-variable relationships are from a General Additive Model (GAM), and gray shading represents the standard error based on a 95% confidence interval. a) Relationship between mean yield and management variation for 21,264 fields in 2018. b) Relationship between yield and environmental variation for 406,570 sampled 30 m pixels between 2008 and 2018.

bound of achievable pixel-level accuracy given the yield monitor dataset, since an extensive machine learning application might explore more sophisticated models and alternative data features that boost accuracy. The SCYM approach was able to account for about 70% of the variation captured in this calibrated model without ground data. We found that the random forests model trained on 1000 fields performed most similarly to SCYM against the test dataset (r^2 = 0.40, RMSE = 2.44 t/ha, MAE = 1.90 t/ha). Accuracy decreased, however, if these 1000 fields were restricted to a single state or year (Fig. 8). Models trained on a single year generalized particularly poorly to all years, with r^2 ranging from 0.10 for the 2014 training sample to 0.35 for the 2017 training sample. This suggests that machine learning approaches will be limited by ground data availability, particularly for historical yield mapping applications that span regional scales and multiple years, demonstrating the value of SCYM's scalable approach.

The preferred SCYM approach also performs well in comparison to other maize yield mapping efforts in the region, particularly in terms of scalability. Gao et al. (2018) fused MODIS and Landsat data to predict yields in 20 Iowa counties from 2001 to 2015, achieving county-level accuracy of r^2 = 0.63. They also found that MODIS alone (r^2 = 0.57) outperformed Landsat alone (r^2 = 0.46), but our results suggest this effect would not hold at sub-field or even field scales. Other promising work to map maize yields in the US Corn Belt has tested a sophisticated data assimilation framework to integrate Landsat observations into crop simulations; testing on selected locations around the US Midwest found field-level accuracies of r^2 = 0.41 and RMSE = 2.17 t/ha (Kang and Özdogan, 2019). At this time, however, the method to downscale to 30 m is computationally intensive and thus was only applied to generate continuous yield maps for select counties in Iowa and Illinois (Kang and Özdogan, 2019); SCYM was able to achieve slightly better field-level performance (r^2 = 0.45, RMSE = 1.85 t/ha) in a more scalable manner, although these two approaches are evaluated on different datasets with different spatial support. Ultimately, SCYM's scalability and sole reliance on Landsat opens the possibility of reconstructing historic yields back to the 1980s across the region, whereas reliance on MODIS or Sentinel restricts efforts to after their launches in 2000 and 2015, respectively.

4.3. Sensitivity to management and environmental factors

Overall, yield estimates from the preferred SCYM approach tracked well with ground yields across underlying variation in field management, soil properties, and annual weather, although some discrepancies and reduced sensitivity are apparent (Fig. 9). SCYM's response to these yield-constraining factors is notable, particularly since no information related to management or soil properties is included in the SCYM yield regression, nor are the weather variables examined here included directly. This both improves confidence in SCYM accuracy and supports the use of satellite approaches such as SCYM for yield gap analyses and evaluation of alternative agronomic management.

4.3.1. Management factors

Based on management data from 21,264 fields in 2018, we found that SCYM and yield monitor ground data responded similarly to planting density and planting date, but SCYM was relatively insensitive to yield impacts from harvest timing (Fig. 9a). Both SCYM and ground

data displayed a positive relationship between yield and seeding rate, with SCYM being only slightly less sensitive than ground data to yield increases as planting density increased (change in GAM-modeled yield predictions between the 5th and 95th percentiles of seeding rate, Δ5–95 = 1.05 vs 1.30 t/ha, respectively). Similarly, both datasets demonstrated that yield decreased with later planting dates, although the penalty for later planting date was smaller in SCYM estimates (Δ5–95 = −1.80 vs −3.36 t/ha). SCYM was most insensitive to yield impacts from harvest dates, with no clear yield penalties from late harvest as seen in the ground data (Δ5–95 = 1.02 vs −1.00 t/ha). This is likely because end-of-season information does not go into SCYM yield estimations and suggests more generally that SCYM would not detect yield impacts from late-season crop disturbances.

4.3.2. Soil and weather factors

Based on the 406,570 pixels across 2008–2018 in the test accuracy dataset, both SCYM and ground data responded similarly to indicators of soil quality (Fig. 9b). Peak yields were observed at intermediate levels of soil available water content, with yield declines in soils that have low water holding capacity as well as soils with high AWC. SCYM also captured a nonlinear response to early season (May) soil moisture, where peak yields were achieved at intermediate levels. Yields generally increased in both datasets as the NCCPI Soil Productivity Index increased, indicating higher yield performance on soils rated more favorable for growing maize. Similar to management variables, however, we found reduced sensitivity in SCYM's response (Δ5–95 = 1.21 vs 2.51 t/ha). SCYM's ability to detect yield variation due to underlying soil properties, while muted, is notable, given that no soil information is used in yield estimation, and provides evidence that it is able to pick up the influences of soil quality on vegetation health through satellite-observed GCVI.

SCYM also showed generally good agreement with ground yields across annual weather variation (Fig. 9b). The preferred SCYM model does include four weather covariables (section 2.2). Here, we examined three weather indicators that are similar but not directly included in the SCYM yield regression. We found that peak yields occurred at intermediate levels of growing season rainfall (May - August) based on both SCYM and ground data. Yield declined as rainfall fell below about 450 mm, indicating water limiting conditions, and when rainfall was above ~800 mm, indicating water-logged conditions. This pattern agrees with recent work finding comparative yield losses from drought and excessive rainfall in the US (Li et al., 2019a). Compared to the ground data, however, we found that SCYM overly penalized yields in low precipitation conditions (e.g., yields at the 5th percentile of rainfall = 4.68 vs 8.21 t/ha) and underpenalized yields in very wet conditions (e.g., yields at the 95th percentile = 10.3 vs 9.00 t/ha). This likely stems from the inclusion of June–August precipitation as a piecewise linear function in the SCYM regression (section 3.2), where non-limiting amounts of precipitation zero out the precipitation term and thus do not affect yields, although yield impacts of wet years have likewise challenged statistical yield predictions at the county level even when more detailed precipitation terms are included due to the complex mechanisms of impact (Li et al., 2019b).

For heat and drought stressors, SCYM estimates captured large yield declines due to high temperatures and high climate water deficit, although in this case SCYM was overly sensitive to these factors compared to the ground data (Δ5–95 for June–August mean temperate = −7.96 vs −4.92 t/ha; Δ5–95 for July climate water deficit = −6.95 vs −4.4 t/ha, Fig. 8b).

5. Conclusions

Retrospective yield mapping provides the opportunity to understand trends and drivers of crop yields over time, a valuable tool for informing best management practices and guiding future adaptation efforts. By utilizing a regionally comprehensive ground dataset spanning 11 years (2008–2018) across the US Corn Belt, we found that our scalable approach can effectively reconstruct historic yields using only Landsat satellite data, explaining 40% of yield variation at the 30 m pixel level and 69% of yield variation at the county level. These scale-dependent measures emphasize the importance of quantifying accuracy at the spatial resolution of the yield map product to best support informed and appropriate interpretation. The ability of SCYM to capture 70% of the variation explained by the ground-calibrated machine learning model demonstrates that region-specific crop simulations can train effective models in the absence of abundant ground data. Comparisons of SCYM yield estimates and ground data along variation in management and biophysical characteristics support the use of satellite-derived yield maps for analyzing yield gaps and management practices. Table 4 summarizes our key recommendations.

Table 4
Key lessons for satellite yield mapping.

Key Lessons	Supporting results
Accuracy metrics from coarse scale aggregations cannot be inferred to apply to smaller field-level and pixel-level scales	Fig. 7
Smoothing satellite time series with harmonic regressions improves accuracy and flexibility	Fig. 5
Capturing crop phenology for effective maize yield mapping requires at least 8 satellite observations per year, including at least one observation during the peak growing season	Fig. 6a
Weather data is most useful when satellite observations are sparse	Table 1
Satellite-derived yield maps are able to detect soil, management, and weather influences	Fig. 9
Satellite-derived yield maps are likely to miss end-of-season weather events	Fig. 9
Ground-calibrated machine learning models may perform poorly when applied to years and regions outside of their training sample	Figure 8:

Data availability

Ground data used in this study are under a strict data use agreement to preserve privacy and are not available. All other input data sources can be acquired from their original source as noted in dataset references and descriptions. Annual yield maps aggregated to 10 km resolution are publicly available on Zenodo at https://doi.org/10.5281/zenodo.4267589. Due to the sensitive nature of high-resolution yield data, the SCYM 30 m yield maps generated by this study are not publicly available for general use. Interested users for non-commercial research purposes may contact David Lobell at dlobell@stanford.edu to arrange a formal data use agreement and access yield maps.

Author credit

JMD, RP, and DBL conceived the paper. RP and SL processed the ground truth dataset. JMD & WD performed the analysis. JMD, RP, SL, WD, and DBL wrote the paper.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank George Azzari for initial algorithm development, Stefania Di Tomasso for assistance with soil and satellite data processing, and Zhenong Jin for initial work with the APSIM simulations. Jake Campolo, Andrew Deines, Stefania Di Tomasso, Hemant Pullabhotla, Matthieu Stigler, Sherrie Wang, and four anonymous reviewers provided helpful

comments on the manuscript. We are grateful to the Google Earth Engine team for continued excellent data processing and analysis infrastructure. Funding was provided by the NASA Harvest Consortium (NASA Applied Sciences Grant No. 80NSSC17K0652, sub-award 54308-Z6059203 to DBL). Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of NASA.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2020.112174.

References

ESA, 2020. Sentinel-2 MSI User Guide. WWW Document.
Abatzoglou, J.T., 2013. Development of gridded surface meteorological data for ecological applications and modelling. Int. J. Climatol. 33, 121–131. https://doi.org/10.1002/joc.3413.
Abatzoglou, J.T., Dobrowski, S.Z., Parks, S.A., Hegewisch, K.C., 2018. TerraClimate, a high-resolution global dataset of monthly climate and climatic water balance from 1958-2015. Sci. Data 5, 1–12. https://doi.org/10.1038/sdata.2017.191.
Azzari, G., Lobell, D.B., 2016. Landsat-based classification in the cloud: an opportunity for a paradigm shift in land cover monitoring. Remote Sens. Environ. https://doi.org/10.1016/j.rse.2017.05.025.
Azzari, G., Jain, M., Lobell, D.B., 2017. Towards fine resolution global maps of crop yields: testing multiple methods and satellites in three countries. Remote Sens. Environ. https://doi.org/10.1016/j.rse.2017.04.014.
Becker-Reshef, I., Justice, C., Sullivan, M., Vermote, E., Tucker, C., Anyamba, A., Small, J., Pak, E., Masuoka, E., Schmaltz, J., Hansen, M., Pittman, K., Birkett, C., Williams, D., Reynolds, C., Doorn, B., 2010. Monitoring global croplands with coarse resolution earth observations: the global agriculture monitoring (GLAM) project. Remote Sens. 2, 1589–1609. https://doi.org/10.3390/rs2061589.
Bivand, R., Keitt, T., Rowlingson, B., 2014. rgdal: Bindings for the Geospatial Data Abstraction Library [WWW Document]. URL. http://cran.r-project.org/package=rgdal.
Blackmore, B.S., Moore, M., 1999. Remedial correction of yield map data. Precis. Agric. 1, 53–66. https://doi.org/10.1023/A:1009969601387.
Breiman, L., 2001. Random forests. Mach. Learn. 45, 5–32. https://doi.org/10.1023/A:1010933404324.
Burke, M., Lobell, D.B., 2017. Satellite-based assessment of yield variation and its determinants in smallholder African systems. Proc. Natl. Acad. Sci. 114, 2189–2194. https://doi.org/10.1073/pnas.1616919114.
Carletto, C., Jolliffe, D., Banerjee, R., 2015. From tragedy to renaissance: improving agricultural data for better policies. J. Dev. Stud. 51, 133–148. https://doi.org/10.1080/00220388.2014.968140.
Cohen, A.A.B., Seifert, C.A., Azzari, G., Lobell, D.B., 2019. Rotation effects on corn and soybean yield inferred from satellite and field-level data. Agron. J. 111, 1–9. https://doi.org/10.2134/agronj2019.03.0157.
R Core Team, 2014. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.
Daughtry, C.S.T., Gallo, K.P., Bauer, M.E., 1983. Spectral estimates of solar radiation intercepted by corn canopies. Agron. J. 75, 527–531.
Deines, J.M., Wang, S., Lobell, D.B., 2019. Satellites reveal a small positive yield effect from conservation tillage across the US Corn Belt. Environ. Res. Lett. 14, 124038. https://doi.org/10.1088/1748-9326/ab503b.
Diker, K., Heermann, D.F., Brodahl, M.K., 2004. Frequency analysis of yield for delineating yield response zones. Precis. Agric. 5, 435–444. https://doi.org/10.1007/s11119-004-5318-9.
Doraiswamy, P., Hatfield, J.L., Jackson, T., Akhmedov, B., Prueger, J., Stern, A., 2004. Crop condition and yield simulations using Landsat and MODIS. Remote Sens. Environ. 92, 548–559. https://doi.org/10.1016/j.rse.2004.05.017.
Fang, H., Liang, S., Hoogenboom, G., 2011. Integration of MODIS LAI and vegetation index products with the CSM-CERES-maize model for corn yield estimation. Int. J. Remote Sens. 32, 1039–1065. https://doi.org/10.1080/01431160903505310.
Franch, B., Vermote, E.F., Skakun, S., Roger, J.C., Becker-Reshef, I., Murphy, E., Justice, C., 2019. Remote sensing based yield monitoring: application to winter wheat in United States and Ukraine. Int. J. Appl. Earth Obs. Geoinf. 76, 112–127. https://doi.org/10.1016/j.jag.2018.11.012.
Fulton, J., Hawkins, E., Taylor, R., Franzen, A., 2018. Yield monitoring and mapping. In: Shannon, K., Clay, D.E., Kitchen, N.R. (Eds.), Precision Agriculture Basics. ASA, CSA, and SSSA, Madison, WI. https://doi.org/10.2134/precisionagbasics.2016.0089.
Gao, F., Anderson, M., Daughtry, C., Johnson, D., 2018. Assessing the variability of corn and soybean yields in Central Iowa using high spatiotemporal resolution multi-satellite imagery. Remote Sens. 10 https://doi.org/10.3390/rs10091489.
Gitelson, A.A., Gritz, Y., Merzlyak, M.N., 2003. Relationships between leaf chlorophyll content and spectral reflectance and algorithms for non-destructive chlorophyll assessment in higher plant leaves. J. Plant Physiol. 160, 271–282. https://doi.org/10.1078/0176-1617-00887.
Gitelson, A.A., Viña, A., Ciganda, V., Rundquist, D.C., Arkebauer, T.J., 2005. Remote estimation of canopy chlorophyll content in crops. Geophys. Res. Lett. 32, 1–4. https://doi.org/10.1029/2005GL022688.
Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore, R., 2017. Google earth engine : planetary-scale geospatial analysis for everyone. Remote Sens. Environ. 202, 18–27. https://doi.org/10.1016/j.rse.2017.06.031.
Green, T.R., Kipka, H., David, O., McMaster, G.S., 2018. Where is the USA Corn Belt, and how is it changing? Sci. Total Environ. 618, 1613–1618. https://doi.org/10.1016/j.scitotenv.2017.09.325.
Guan, K., Wu, J., Kimball, J.S., Anderson, M.C., Frolking, S., Li, B., Hain, C.R., Lobell, D.B., 2017. The shared and unique values of optical, fluorescence, thermal and microwave satellite data for estimating large-scale crop yields. Remote Sens. Environ. 199, 333–349. https://doi.org/10.1016/j.rse.2017.06.043.
Hijmans, R.J., 2014. raster: Geographic data analysis and modeling.
Hoffman, L.A., Etienne, X.L., Irwin, S.H., Colino, E.V., Toasa, J.I., 2015. Forecast performance of WASDE price projections for U.S. corn. Agric. Econ. (United Kingdom) 46, 157–171. https://doi.org/10.1111/agec.12204.
Holzworth, D.P., Huth, N.I., deVoil, P.G., Zurcher, E.J., Herrmann, N.I., McLean, G., Chenu, K., van Oosterom, E.J., Snow, V., Murphy, C., Moore, A.D., Brown, H., Whish, J.P.M., Verrall, S., Fainges, J., Bell, L.W., Peake, A.S., Poulton, P.L., Hochman, Z., Thorburn, P.J., Gaydon, D.S., Dalgliesh, N.P., Rodriguez, D., Cox, H., Chapman, S., Doherty, A., Teixeira, E., Sharp, J., Cichota, R., Vogeler, I., Li, F.Y., Wang, E., Hammer, G.L., Robertson, M.J., Dimes, J.P., Whitbread, A.M., Hunt, J., van Rees, H., McClelland, T., Carberry, P.S., Hargreaves, J.N.G., MacLeod, N., McDonald, C., Harsdorf, J., Wedgwood, S., Keating, B.A., 2014. APSIM - evolution towards a new generation of agricultural systems simulation. Environ. Model. Softw. 62, 327–350. https://doi.org/10.1016/j.envsoft.2014.07.009.
Hunt, M.L., Blackburn, G.A., Carrasco, L., Redhead, J.W., Rowland, C.S., 2019. High resolution wheat yield mapping using Sentinel-2. Remote Sens. Environ. 233, 111410. https://doi.org/10.1016/j.rse.2019.111410.
Jain, M., Balwinder-Singh Rao, P., Srivastava, A.K., Poonia, S., Blesh, J., Azzari, G., McDonald, A.J., Lobell, D.B., 2019. The impact of agricultural interventions can be doubled by using satellite data. Nat. Sustain. 2, 931–934. https://doi.org/10.1038/s41893-019-0396-x.
Jain, M., Srivastava, A., Balwinder-Singh Joon, R., McDonald, A., Royal, K., Lisaius, M., Lobell, D., 2016. Mapping smallholder wheat yields and sowing dates using micro-satellite data. Remote Sens. 8, 860. https://doi.org/10.3390/rs8100860.
Jin, Z., Zhuang, Q., Tan, Z., Dukes, J.S., Zheng, B., Melillo, J.M., 2016. Do maize models capture the impacts of heat and drought stresses on yield? Using algorithm ensembles to identify successful approaches. Glob. Chang. Biol. 22, 3112–3126. https://doi.org/10.1111/gcb.13376.
Jin, Z., Azzari, G., Lobell, D.B., 2017. Improving the accuracy of satellite-based high-resolution yield estimation: a test of multiple scalable approaches. Agric. For. Meteorol. 247, 207–220. https://doi.org/10.1016/j.agrformet.2017.08.001.
Jin, Z., Azzari, G., You, C., Di Tommaso, S., Aston, S., Burke, M., Lobell, D.B., 2019. Smallholder maize area and yield mapping at national scales with Google earth engine. Remote Sens. Environ. 228, 115–128. https://doi.org/10.1016/j.rse.2019.04.016.
Kamir, E., Waldner, F., Hochman, Z., 2020. Estimating wheat yields in Australia using climate records, satellite image time series and machine learning methods. ISPRS J. Photogramm. Remote Sens. 160, 124–135. https://doi.org/10.1016/j.isprsjprs.2019.11.008.
Kang, Y., Özdogan, M., 2019. Field-level crop yield mapping with Landsat using a hierarchical data assimilation approach. Remote Sens. Environ. 228, 144–163. https://doi.org/10.1016/j.rse.2019.04.005.
Kayad, A., Sozzi, M., Gatto, S., Marinello, F., Pirotti, F., 2019. Monitoring within-field variability of corn yield using sentinel-2 and machine learning techniques. Remote Sens. 11 https://doi.org/10.3390/rs11232873.
Kharel, T.P., Swink, S.N., Maresma, A., Youngerman, C., Kharel, D., Czymmek, K.J., Ketterings, Q.M., 2019. Yield monitor data cleaning is essential for accurate corn grain and silage yield determination. Agron. J. 111, 509–516. https://doi.org/10.2134/agronj2018.05.0317.
Kholsa, R., Flynn, B., 2008. Understanding and cleaning yield monitor data. In: Logsdon, S., Clay, D., Moore, D., Tsegaya, T. (Eds.), Soil Science: Step-by-Step Field Analysis. Soil Science Society of America, Madison, WI, pp. 113–130. https://doi.org/10.2136/2008.soilsciencestepbystep.c9.
Lai, Y.R., Pringle, M.J., Kopittke, P.M., Menzies, N.W., Orton, T.G., Dang, Y.P., 2018. An empirical model for prediction of wheat yield, using time-integrated Landsat NDVI. Int. J. Appl. Earth Obs. Geoinf. 72, 99–108. https://doi.org/10.1016/j.jag.2018.07.013.
Lambert, M.J., Traoré, P.C.S., Blaes, X., Baret, P., Defourny, P., 2018. Estimating smallholder crops production at village level from Sentinel-2 time series in Mali's cotton belt. Remote Sens. Environ. 216, 647–657. https://doi.org/10.1016/j.rse.2018.06.036.
Li, Y., Guan, K., Schnitkey, G.D., Delucia, E., Peng, B., 2019a. Excessive rainfall leads to maize yield loss of a comparable magnitude to extreme drought in the United States. Glob. Chang. Biol. 1–13 https://doi.org/10.1111/gcb.14628.
Li, Y., Guan, K., Yu, A., Peng, B., Zhao, L., Li, B., Peng, J., 2019b. Toward building a transparent statistical model for improving crop yield prediction: modeling rainfed corn in the U.S. F. Crop. Res. 234, 55–65. https://doi.org/10.1016/j.fcr.2019.02.005.
Liaw, A., Wiener, M., Breiman, L., Cutler, A., 2002. Classification and regression by randomForest. R News 2, 18–22.
Liu, L., Basso, B., 2020. Linking field survey with crop modeling to forecast maize yield in smallholder farmers' fields in Tanzania. Food Secur. https://doi.org/10.1007/s12571-020-01020-3.
Lobell, D.B., 2013. The use of satellite data for crop yield gap analysis. F. Crop. Res. 143, 56–64. https://doi.org/10.1016/j.fcr.2012.08.008.

Lobell, D.B., Azzari, G., 2017. Satellite detection of rising maize yield heterogeneity in the U.S. Midwest. Environ. Res. Lett. 12, 014014. https://doi.org/10.1088/1748-9326/aa5371.

Lobell, D.B., Cassman, K.G., Field, C.B., 2009. Crop yield gaps: their importance, magnitudes, and causes. Annu. Rev. Environ. Resour. 34, 179–204. https://doi.org/10.1146/annurev.environ.041008.093740.

Lobell, D.B., Thau, D., Seifert, C., Engle, E., Little, B., 2015. A scalable satellite-based crop yield mapper. Remote Sens. Environ. 164, 324–333. https://doi.org/10.1016/j.rse.2015.04.021.

Lobell, D.B., Di Tommaso, S., You, C., Yacoubou Djima, I., Burke, M., Kilic, T., 2019. Sight for sorghums: comparisons of satellite- and ground-based Sorghum yield estimates in Mali. Remote Sens. 12, 100. https://doi.org/10.3390/rs12010100.

Luck, J.D., Precision, E., Engineer, A., 2014. Precision Agriculture Best Management Practices for Collecting Accurate Yield Data and Avoiding Errors during Harvest, EC2004. Lincoln, NE.

MacDonald, R.B., Hall, F.G., 1980. Global crop forecasting. Science (80-.) 208, 670–679. https://doi.org/10.1126/science.208.4445.670.

Maestrini, B., Basso, B., 2018. Drivers of within-field spatial and temporal variability of crop yield across the US Midwest. Sci. Rep. 8, 1–9. https://doi.org/10.1038/s41598-018-32779-3.

Mueller, S.M., Messina, C.D., Vyn, T.J., 2019. Simultaneous gains in grain yield and nitrogen efficiency over 70 years of maize genetic improvement. Sci. Rep. 9, 1–8. https://doi.org/10.1038/s41598-019-45485-5.

NASS, 2019. Quick Stats API [WWW Document]. USDA Natl. Agric. Stat. Serv. https://quickstats.nass.usda.gov/api (accessed 6.1.19).

Nguy-Robertson, A., Gitelson, A., Peng, Y., Viña, A., Arkebauer, T., Rundquist, D., 2012. Green leaf area index estimation in maize and soybean: combining vegetation indices to achieve maximal sensitivity. Agron. J. 104, 1336–1347. https://doi.org/10.2134/agronj2012.0065.

NRCS, 2016. SSURGO Web Soil Survey. USDA Nat. Recources Conserv. Serv.

Pebesma, E., 2018. Simple features for R: Standardized support for spatial vector data. R J. 10, 439–446. https://doi.org/10.32614/rj-2018-009.

Roy, D.P., Yan, L., 2018. Robust Landsat-based crop time series modelling. Remote Sens. Environ. 0–1. https://doi.org/10.1016/j.rse.2018.06.038.

Seifert, C.A., Azzari, G., Lobell, D.B., 2018. Satellite detection of cover crops and their effects on crop yield in the Midwestern United States. Environ. Res. Lett. 13, 064033 https://doi.org/10.1088/1748-9326/aac4c8.

Simbahan, G.C., Dobermann, A., Ping, J.L., 2004. Screening yield monitor data improves grain yield maps. Agron. J. 96, 1091–1102. https://doi.org/10.2134/agronj2004.1091.

Skakun, S., Vermote, E., Franch, B., Roger, J.C., Kussul, N., Ju, J., Masek, J., 2019. Winter wheat yield assessment from Landsat 8 and Sentinel-2 data: incorporating surface reflectance, through phenological fitting, into regression yield models. Remote Sens. 11 https://doi.org/10.3390/rs11151768.

Slater, A.G., 2016. Surface solar radiation in North America: a comparison of observations, reanalyses, satellite, and derived products. J. Hydrometeorol. 17, 401–420. https://doi.org/10.1175/JHM-D-15-0087.1.

Sun, W., Whelan, B., McBratney, A.B., Minasny, B., 2013. An integrated framework for software to provide yield data cleaning and estimation of an opportunity index for site-specific crop management. Precis. Agric. 14, 376–391. https://doi.org/10.1007/s11119-012-9300-7.

Thornton, P.E., Thornton, M.M., Mayer, B.W., Wei, Y., Devarakonda, R., Vose, R.S., Cook, R.B., 2016. Daymet: Daily Surface Weather Data on a 1-Km Grid for North America, Version 3. https://doi.org/10.3334/ORNLDAAC/1328.

USGS, 2017. Landsat Collection 1 Level 1 Product Definition Version 1.0.

Van Ittersum, M.K., Cassman, K.G., Grassini, P., Wolf, J., Tittonell, P., Hochman, Z., 2013. Yield gap analysis with local to global relevance-a review. F. Crop. Res. 143, 4–17. https://doi.org/10.1016/j.fcr.2012.09.009.

Viña, A., Gitelson, A.A., Nguy-Robertson, A.L., Peng, Y., 2011. Comparison of different vegetation indices for the remote assessment of green leaf area index of crops. Remote Sens. Environ. 115, 3468–3478. https://doi.org/10.1016/j.rse.2011.08.010.

Vogelmann, J.E., Gallant, A.L., Shi, H., Zhu, Z., 2015. Perspectives on monitoring gradual change across the continuity of Landsat sensors using time-series data. Remote Sens. Environ. https://doi.org/10.1016/j.rse.2016.02.060.

Waldner, F., Horan, H., Chen, Y., Hochman, Z., 2019. High temporal resolution of leaf area data improves empirical estimation of grain yield. Sci. Rep. 9, 15714. https://doi.org/10.1038/s41598-019-51715-7.

Wang, S., Azzari, G., Lobell, D.B., 2019. Crop type mapping without field-level labels: random forest transfer and unsupervised clustering techniques. Remote Sens. Environ. 222, 303–317. https://doi.org/10.1016/j.rse.2018.12.026.

Weiss, M., Jacob, F., Duveiller, G., 2020. Remote sensing for agricultural applications: a meta-review. Remote Sens. Environ. 236, 111402. https://doi.org/10.1016/j.rse.2019.111402.

Whitcraft, A.K., Becker-Reshef, I., Justice, C.O., 2015. A framework for defining spatially explicit earth observation requirements for a global agricultural monitoring initiative (GEOGLAM). Remote Sens. 7, 1461–1481. https://doi.org/10.3390/rs70201461.

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T., Miller, E., Bache, S., Müller, K., Ooms, J., Robinson, D., Seidel, D., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K., Yutani, H., 2019. Welcome to the tidyverse. J. Open Source Softw. 4, 1686. https://doi.org/10.21105/joss.01686.

Wolanin, A., Mateo-Garciá, G., Camps-Valls, G., Gómez-Chova, L., Meroni, M., Duveiller, G., Liangzhi, Y., Guanter, L., 2020. Estimating and understanding crop yields with explainable deep learning in the Indian Wheat Belt. Environ. Res. Lett. 15 https://doi.org/10.1088/1748-9326/ab68ac.

Wood, S.N., 2017. Generalized Additive Models: An Introduction with R, 2nd ed. Chapman and Hall.

Wulder, M.A., Coops, N.C., Roy, D.P., White, J.C., Hermosilla, T., 2018. Land cover 2.0. Int. J. Remote Sens. 39, 4254–4284. https://doi.org/10.1080/01431161.2018.1452075.

Yan, L., Roy, D.P., 2016. Conterminous United States crop field size quantification from multi-temporal Landsat data. Remote Sens. Environ. 172, 67–86. https://doi.org/10.1016/j.rse.2015.10.034.

Zhu, P., Zhuang, Q., Archontoulis, S.V., Bernacchi, C., Müller, C., 2019. Dissecting the nonlinear response of maize yield to high temperature stress with model-data integration. Glob. Chang. Biol. 25, 2470–2484. https://doi.org/10.1111/gcb.14632.

Zipper, S.C., Soylu, M.E., Booth, E.G., Loheide, S.P., 2015. Untangling the effect of shallow groundwater and soil texture as drivers of subfield-scale yield variability. Water Resour. Res. 51, 1–21. https://doi.org/10.1002/2015WR017522.

Zipper, Samuel C, Stack Whitney, Kaitlin, Deines, Jillian M, Befus, Kevin M, Bhatia, Udit, Albers, Sam J, Beecher, Janice, Brelsford, Christa, Garcia, Margaret, Gleeson, Tom, O'Donnell, Francis, Resnik, David, Schlager, Edella, et al., 2019. Balancing open science and data privacy in the water sciences. Water Resources Research. https://doi.org/10.1029/2019WR025080.